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THESIS

Virtual Reality Simulation for Fire Safety Emergency Training

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1. INTRODUCTION

1.1 Background and Relevance of Fire Safety Training

Fire safety is widely regarded as an important part of education and workplace training. A fire can spread very quickly, and within only a few minutes it can cause serious property damage and, more importantly, threaten human life [1]. For this reason schools, universities and companies are expected to provide fire safety training. In some cases, this is simply best practice, and in some cases it is required by law. Yet, despite its importance, the way fire safety is taught and the effectiveness of such training is rarely as beneficial as intended.

In the European Union alone, there is an estimated over 150,000 building fires every year leading to approximately 3500 fatalities and close to 20,000 injuries [2]. Data provided by Hungarian National Directorate of Disaster Management shows that in 2022, more than 8000 structural fires were reported in Hungary, and majority of them were in residential or public buildings [3]. These statistics are a clearly indicate that fire safety training is not just a formality but it is a huge factor when it comes to disaster prevention, that could save lives and avoid damage to property provided it is done properly.

The most common approach is a traditional fire drill. Drills are valuable in many ways. They familiarize participants with escape routes and show the panic of rush during evacuations, and they remind people of the seriousness of fire emergencies. At the same time, drills are not perfect. In large schools or universities, drills may involve many hundreds or even thousands of students moving at once and what is supposed to be a planned exercise can quickly become a very stressful event. Other factors like crowded corridors, possible harsh weather either hot or cold, and the general confusion of being part of a large moving group all add to the challenge. According to reports from the European Fire Safety Alliance, many drill-related incidents in schools involve mild panic and sometimes even injury due to overcrowding [4]. While these cases are usually a select few, they show that drills, which are meant to prepare people, sometimes also create unnecessary stress and strain.

Besides fire drills and evacuations, demonstrations on how to handle fire safety equipment is a much a more practical and important aspect of safety education. Although, it is not possible to give every student or employee a meaningful hands-on training in one single drill because a school cannot realistically let each and every student practice with an extinguisher, and a company cannot pause their business operations long enough for every new worker to rehearse

detailed emergency procedures. Research studies conducted by the National Fire Protection Association (NFPA) reflect that over 60 percent of participants forget a lot of important evacuation steps in a six months period after a single drill demonstrating the importance of practical methods when it comes to long-term memory [5]. As a result, fire safety training often ends up reduced to lectures, short video presentations or one-time demonstrations. These methods are very easy to arrange, but they only provide surface-level knowledge of what to do when faced with a real fire.

The effectiveness of this training is also be affected by age and maturity. Younger children are easily intimidated by fire drills so instead of learning them they might not take away the lessons in the manner it was presented and whereas older learners, like those in high school or university, are much more likely to take the training about fire safety more seriously. In workplaces, fire safety is often covered during the employee onboarding process, when employees are still adapting and open to new information. Even then, the training they receive is usually limited to just a quick orientation. Research has shown that individuals who go through refreshers on safety policies, retain memory about these procedures significantly better during high-stress situations, which supports the idea that one time training is not enough to build an automatic response in everyone [6].

Across all conventional methods, the greatest limitation is realism. The conditions of a true fire cannot be reproduced without recreating danger. Smoke, noise, confusion and the pressure to make quick choices, these are all crucial elements of fire emergencies but they cannot be safely included in practice drills. What participants experience is a watered-down version of reality, which is useful to some extent, but still incomplete. Previous safety studies note that most participants behave very differently in simulations as compared to real emergencies, mainly because there is no emotional pressure [7].

In recent years, alternative approaches like Virtual Reality (VR) have been considered and has emerged as one of the most promising technologies in education, compared to the drills, VR actually exposes the participants by placing them inside real life situations without any risks involved at all. It also allows for repetition, which has proven to be essential for learning. Additionally this method helps avoid the costs and logistics of having to organizing full-scale exercises repeatedly in large institutions or organizations. Schools and workplaces can both use VR as a complement to their traditional means which can offer a more flexible way to practice.

A number of trials conducted in educational and corporate environments found that participants who trained using VR were able to recall escape routes better and find the way out quickly and make more confident decisions during these simulated emergencies [8]. A VR-based system is not meant to replace real-world drills entirely but attempts to build on them by filling in the gaps it leaves behind and improving the overall experience. The result, at least in principle is a training experience that is more consistent and more engaging to the user making it potentially more effective in the long run. By exploring how such systems can be introduced into fire safety training, this thesis aims to show that immersive technologies can support modern fire safety education.

1.2 Problem Statement and Motivation

Fire safety training is essential but it is also often limited in its current form. Traditional exercises which are common, face logistical challenges especially in large institutions. As a result, many drills end up being a box to tick rather than being practically understood because they raise awareness but do not always prepare people to make quick and confident decisions during an actual emergency. In many large scale evaluations, less than 40 percent of participants were able to recall evacuation steps correctly after a single annual drill, which points out that the one time practice does not build long term memory neither does it train participants to react with their instincts [9].

These weaknesses leave a gap between what is supposed to be achieved through fire safety education and the realistic outcome for participants after such scenarios. Furthermore, younger learners may tend to associate drills with fear or confusion, while older learners and employees often treat them just as a formality and take it in an unserious manner. Psychological studies done on learning indicate that fear based drills or purely procedural drills can effect skill retention because learners associate the event with discomfort instead of understanding [10].

This study addresses this gap by introducing a Virtual Reality (VR) based training system which is designed to enhance practices by making them repeatable. Instead of viewing VR as a replacement for existing methods, this approach uses it as a supplement that enables active participation and a more consistent evaluation of the training. Similar VR-based safety programs which have been implemented in industries such as aviation, healthcare, and construction have already shown improvement in both recalling procedures and in behavioural performance during simulated emergencies [11].

In addition to enhancing realism, the objective is to make it easier train larger groups of users and also to maintain the consistency of training in these different groups. A single VR setup can be used to train a large number of participants over multiple rounds of training without disturbing normal operations, while ensuring that schools and organizations do not compromise any safety standards. Corporate VR training reports that were produced by PricewaterhouseCoopers (PwC) indicate that it is possible to train employees four times quicker by using immersive systems than in a classroom environment which enhance not only focus but also the confidence in making decisions under high pressure [12]. By focusing on VR's ability in keeping the person engaged and making sure it is accessible to schools and other organizations. This research aims to demonstrate how immersive tools can bridge the gap between how ready we are in theory and the real-world when it comes to fire safety education.

1.3 Objectives of the Study

The objectives of this study are as follows:

- To design and implement an interactive Virtual Reality (VR) simulation that teaches users the basics of fire safety in a safe and controlled environment.
- To build scenarios that depict different types of fire hazards and allow users to practice choosing and using the correct extinguishers.
- To evaluate the potential of VR as a training tool in schools, universities, and workplaces by comparing its advantages with traditional fire drills.
- To establish a foundation for future development of VR fire safety training in more demanding environments, such as high-risk workplaces or firefighter preparation.

Although the current project is not intended to substitute conventional fire drills, it is meant to supplement them with a controlled, repeatable environment to practice in a safe way. It targets practical, first-time learners in schools and in workplaces, or more precisely those groups that tend to receive surface level fire safety education. The study enables participants to practice the necessary procedures and decision-making without the logistical or safety limitations of actual exercises by integrating an interactive learning environment. It also takes into consideration the use of advanced simulations in the future for high-risk professions.

2. VIRTUAL REALITY: CONCEPTS AND FOUNDATIONS

Virtual Reality (VR) is a three-dimensional and computer-generated world that can be explored and interacted with like it is part of the real world [13]. Compared to the traditional two-dimension interfaces, VR immerses the user in a digitally created environment that reacts to head and body movements in real-time. A properly designed VR system can create this illusion of presence, that believable feeling of being inside a different environment by replacing the actual environment of the user with artificial imaging and audio [14].

The fundamental aspects of VR that make this possible are immersion, interaction, and sensory feedback, each of these contribute to the user's overall sense of realism [15].

2.1.1 Immersion

Immersion refers to the extent to how well a virtual world is able to grasp the attention of the user, and replace the physical world perceived by the user [15]. Many things influence this perception, such as the system's field of view, display resolution, frame rate, spatial audio, and getting rid of distractions in the real-world. Some head mounted displays which are high end can increase Immersion along with motion tracking systems that synchronize the virtual visual display precisely with the user's movement, because the illusion strengthens when sensory cues function the way users expect it to.

2.1.2 Interaction

Interaction also transforms VR from being observed passively into active participation. Controllers, gloves or even body tracking suits configured with the VR system, allow users to interact with objects, navigate through spaces and cause events which the system instantly reacts to. Interactions should be designed in a way that makes actions in VR feel natural to the user, by mimicking physical behaviors such as picking up objects, clicking buttons or turning their heads to look around the environment [14]. This continuous feedback keeps users engaged in the simulation and makes them feel mentally involved, which is essential for simulations.

2.1.3 Sensory Feedback

Although the sense of vision is very dominant in most virtual experiences, other sensory mediums can increase realism by combining spatial audio, which gives the user a sense of direction where the sound is coming from and how far away it is, along with haptic feedback

which is a vibration based response to events that happen inside the simulation that mimics things like texture and level of impact through handheld devices or gloves. Some advanced setups include motion platforms and environmental effects such as airflow which can simulate physical sensations like speed or height, these effects can also be temperature controlled making the experience more immersive. One factor that can either make or break the illusion for sensory feedback of this type is synchronizing all these signals with the user's movements, because any mismatch can cause the user discomfort or even break immersion [13].

2.1.4 Presence and Realism

This relation between immersion, interaction and sensory feedback is what creates presence, or the experience of being in the virtual world. Presence is strengthened when system latency is as low as possible and sensory information remains consistent across all the senses involved. The heightened sense of realism is not only more engaging for the user but it also enhances performance, since people tend to react to virtual stimuli in the same way they do to real-life scenarios [15].

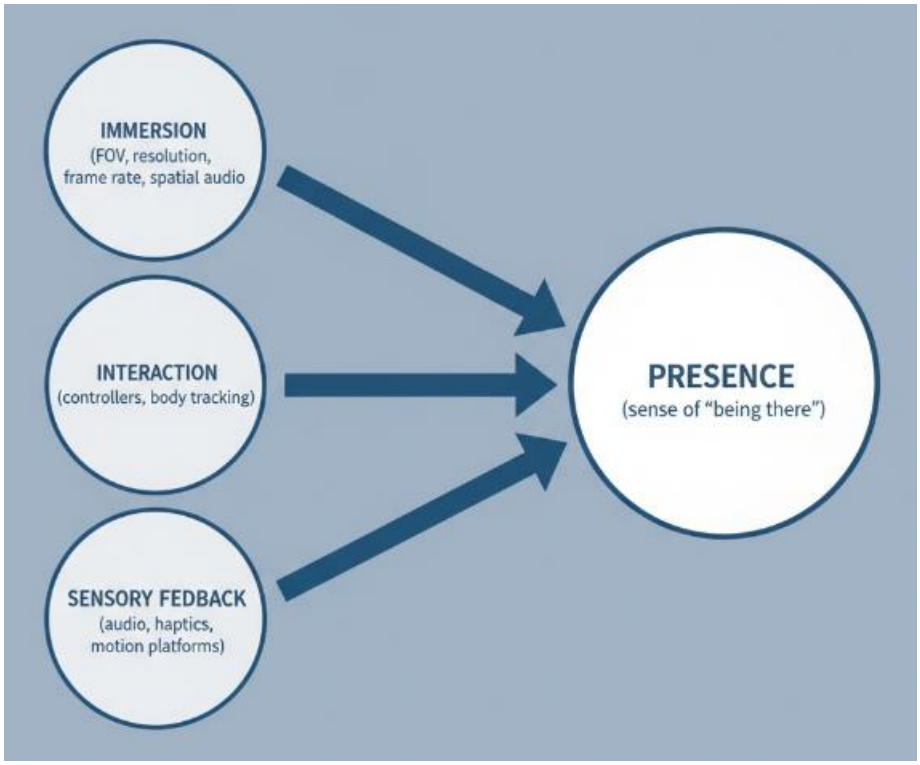

Figure 1: Key Components of Presence in VR

2.1.5 Summary

VR is based on the combination of visuals that are immersive, controls that feel interactive, and sensory feedback that makes simulated environments appear more life like. The balance of these principles in any VR system is what determines the realism.

2.2 Historical Development of VR

2.2.1 Origins and Early Development

Virtual Reality was a notion long before it was possible to create it in a modern computer. The initial ideas of simulated experiences were based on mechanical and optical systems that attempted to reproduce alternate realities for the user. The Sensorama (1956) by Morton Heilig was one of the first to offer a multisensory experience, using stereoscopic images, stereo sound, vibration and even smell to recreate the feeling of a motorbike ride through New York City [16]. It was all mechanical, but it felt and smelled like VR, as it evoked several of the senses in an illusion of presence.

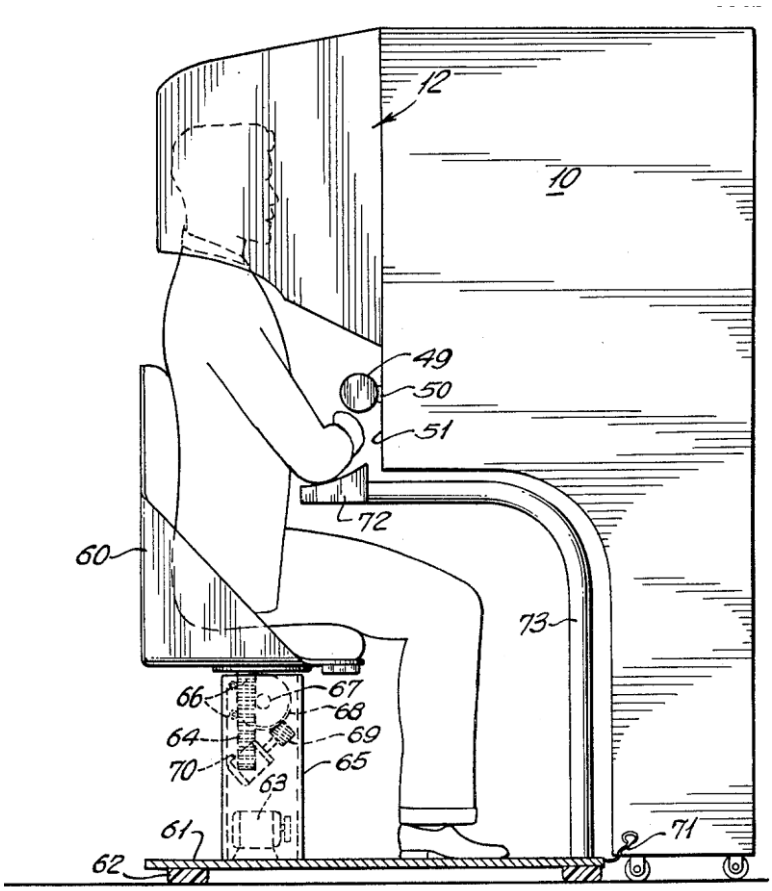

Figure 2 : Patent Diagram of the Sensorama (1956)

Source: Wikimedia Commons (Public Domain), https://commons.wikimedia.org/w/index.php?curid=3616641

The 1960s and 1970s were the years of imagination to the digital world. The Sword of Damocles (1968) by Ivan Sutherland is widely considered to be the first actual head-mounted display (HMD) as it consisted of head tracking and real-time computer-generated graphics to adjust the perspective based on the movements of the user [17]. Over the following decades, the application of VR in training pilots and soldiers was discussed by research institutions and military organizations [18]. Though the systems of this time were hard to manage and limited by processing power, they established the technical foundation for immersive displays, motion tracking, and real-time simulation that we have now.

2.2.2 Expansion and Commercial Experiments

The 1990s represented VR's first significant step into the public. Consumer interest rose when companies like Sega and Nintendo made their first attempts at experimenting with headsets like Sega VR and Virtual Boy [19]. At the same time, the CAVE (Cave Automatic Virtual Environment) was proposed in academic research as a room-scale projection system that enclosed users within stereoscopic images on several walls [20]. Despite the widespread excitement that came with it, these systems faced major challenges, including low display resolution, high latency and high costs, which severely limited their access to the public. As expectations exceeded performance, the VR market declined rapidly toward the end of the decade which led to a significant slowdown in development.

2.2.3 Revival of VR Technology

Despite declining commercial interest, technological advances had been made in silence into the 2000s. The developments in computer graphics, motion sensors and display technologies provided a foundation to a renewed interest in immersive media. High-quality VR was more accessible than ever, with low cost consumer-grade hardware including better motion sensors and high-resolution displays. This technological foundation led to the creation of the Oculus Rift prototype in the early 2010s, which reawakened global excitement and interest in Virtual Reality [21].

The effectiveness of the crowdfunding campaigns and rapid speed of hardware development demonstrated that VR could finally become available to the consumers. By the mid-2010s, headsets such as the Oculus Rift CV1, HTC Vive, and PlayStation VR offered spatially interactive experiences, precise tracking, and interactive hand controllers. These advances have

made virtual reality (VR) more than just a scientific fascination, but a useful tool for simulation, learning and entertainment.

2.2.4 The Modern Era of VR and Mixed Reality

One of the most significant advancements in how VR technology functions was the introduction of standalone headsets like the Oculus Quest that popularized inside out tracking which relied on onboard cameras, meaning they no longer needed a computer or any external devices to effectively track the movement of the user in real-time [22]. This invention made VR more portable and user-friendly, and most importantly, affordable. As newer headset companies began integrating features such as eye and hand tracking as well as haptic controllers to improve the precision of immersion, the competition in the VR world skyrocketed.

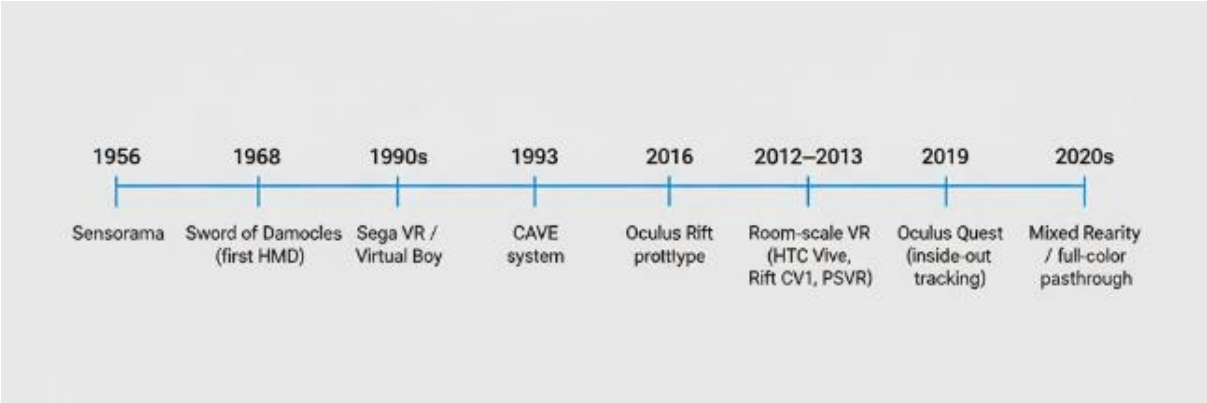

Figure 3 : Timeline of VR Development

In the 2020s, the boundary between Virtual Reality and Mixed Reality (MR) began to blur. The introduction of modern devices that provided full-color passthrough gave users the ability to see and interact with both the physical and the virtual worlds at the same time [23]. This shift took VR beyond the world of entertainment, finding new purpose in classrooms, hospitals, design studios, and professional training environments. Companies began using VR setups and simulators in order to teach safety procedures, allow people to work together from different locations, and help individuals develop hands-on expertise in safe environments.

While a few challenges of motion sickness, limited comfort, and accessibility still persist, these advancements have formed the technological foundation of the modern VR system.

2.3 How VR Works: Technical Overview

Virtual Reality (VR) is a system that involves both hardware and software components which interact with each other to create a three-dimensional digital space that reacts to the movements and sensations of a user [24]. The fundamental idea of any VR system is that of real-time interaction where each action or movement taken by the user leads to a change in what the user can see, hear, or feel in the virtual world. This section proceeds to outline the essential elements of a VR system such as tracking, display, input, and rendering and the way they interact to create an immersive experience.

2.3.1 Tracking Systems

Tracking enables VR to react to the user's body movements accurately, it does this through a combination of sensors, cameras, and gyroscopes that constantly calculate and determine the position and orientation of the head, hands, and in some cases even the entire body. There are two primary approaches to this, outside-in tracking and inside-out tracking [25].

Outside-in tracking relies on external sensors being placed around the user's environment within the system rig. This is done by sensors that detect the location of the headset and controllers with the help of either infrared lights (IR) or laser based systems, devices like the HTC Vive and Oculus Rift CV1 typically use this approach. Inside-out tracking on the other hand, uses cameras that are installed in the headset itself to track the environment and the user’s movements without requiring any external equipment. This way, the setup is much easier and it also allows for more freedom while moving, it was popularized by headsets such as the Oculus Quest.

2.3.2 Display Systems

The headset's display system is responsible for projecting the virtual environment to the user's field of view (FOV). VR Headsets have a small display for each eye, and when viewed together, it forms a stereoscopic image that creates the illusion of depth, giving the scene a three dimensional appearance. Additionally, modern displays use higher resolutions and higher refresh rates, which ensure better visuals and less motion blur when the user looks or moves around.

Broadening the field of view (FOV) makes the feeling of immersion more powerful, this can be done if the image is focused then slightly distorted using optical lenses in such a way that at

closer distances, it becomes natural and spacious as opposed to the flat image of a regular display. High-end systems have adjustable lenses and eye tracking to automatically change focus and perspective depending on where the user is looking to make them even more realistic.

2.3.3 Input and Interaction Devices

The virtual world is brought to life by interaction. Handheld controllers, motion trackers, gloves and even full body suits are used as common input devices in advanced setups. These gadgets recognize gestures, button presses and the spatial movement which enables the user to interact directly with the virtual objects.

Many modern controllers include haptic feedback, which produces small vibrations or pressure responses when interacting with objects inside the simulation, conveying the sensation of texture and impact. Some advanced VR setups even include treadmills or motion platforms, these are essentially platforms that let users physically walk in place in either a linear or a multidirectional manner, creating the illusion of moving deeper through a larger virtual world.

Voice recognition, hand tracking and eye tracking are becoming more popular as input devices and eliminate the need to use physical controllers, as well as providing a more natural and seamless user experience.

2.3.4 Rendering and Software Systems

The scene viewed through a VR headset is generated using highly advanced graphics software, which creates images in real time by using the user position and input. Each frame of the virtual scene must update immediately as the user moves, to create the illusion that the virtual world still exists whether or not the person is currently observing it.

In order to achieve this, VR engines, such as Unity or Unreal Engine are equipped with advanced algorithms that process lighting, physics, shadows, and allow playing perspective. They also include motion prediction which estimates the next direction of the user's head movement to minimize the latency and provide a smoother transition [26]. The rendering software continuously communicates with the tracking and input systems to make sure that all the components are completely synchronized.

Spatial audio systems are crucial in simulating the direction and distance of the all the sounds that the scene may have, which enhances the concept of realism because sound can get louder

as you approach the source like footsteps, and voices change naturally based on the direction of the head.

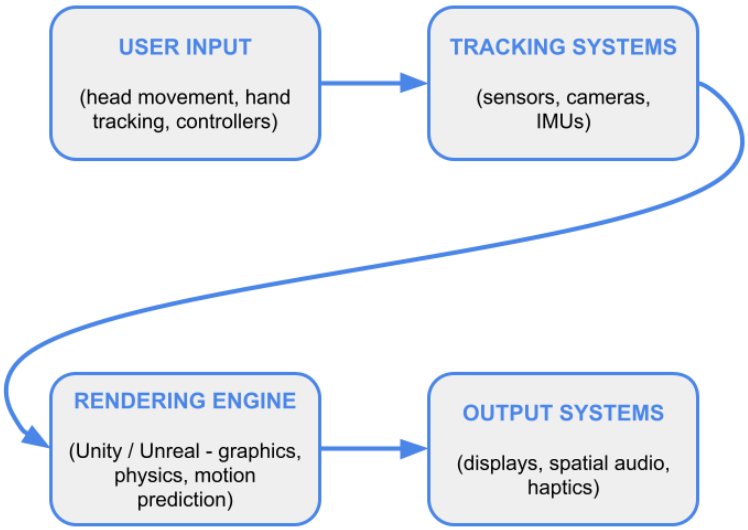

Figure 4 : Basic VR System Pipeline

2.3.5 System Performance and Latency

Performance for VR is usually measured by how well it can handle frame rate and latency. Frame rate determines how many images are being displayed per second. This is responsible for smoother visuals as the higher the rate of frames, the smoother the picture. Whereas, latency measures the delay between a user's action and the corresponding reaction in the simulation, which should ideally be below 20 milliseconds since high latency can lead to nausea or dizziness in addition to breaking the immersion [27].

There are techniques such as 'foveated rendering' where only the area of the image being directly looked at is rendered in full detail, this reduces the load on the computer while running heavy simulations, preserving the quality of the image [27].

2.4 VR Hardware Devices and Interfaces

The bridge between the digital simulation and the user is formed by the hardware used to enable the immersion, each component, be it visual, tactile or auditory, works together to maintain the immersion and realism. These devices need to be precise in order to influence how believable and comfortable they make the simulated experience for the user.

2.4.1 Head-Mounted Displays (HMDs)

The Head Mounted Display or HMD is the key component of any VR system, serving as both the display device and also as the user's point of connection with the virtual world. Each display shows slightly different images to the left and right eye that creates a binocular disparity, this is the same visual mechanism that the brain uses to perceive depth in reality [28]. The result of this combination of offset images creates a convincing sense of three-dimensional space.

These lenses that are inside the headset are made curved to stretch and focus the image, making nearby objects look natural instead of flat. If this feature is not calibrated properly, it causes misalignment and distortion which can make the user motion sick or feel discomfort while looking around.

Sometimes VR simulations can go on for more than just a few minutes, maybe even an hour, so it is important that the headset is designed in an ergonomic fashion to keep it well balanced. This not only minimizes neck strain, but also helps maintain accuracy in tracking as users would feel comfortable to move around naturally without hesitation.

2.4.2 Input Devices and Controllers

VR controllers act as an extension of the user but inside the simulation by converting the users physical hand movements in the real world into the corresponding digital action. These controllers, unlike conventional game controllers, are built in with multiple sensors that analyze the position and orientation in real time. In most systems nowadays, headsets are able to track the controller's location within the three dimensional space using infrared (IR) LEDs or inertial measurement units (IMUs).

Figure 5 : Example of VR Motion Controllers (Oculus Touch)

Source: Samwalton9, Wikimedia Commons, CC BY-SA 4.0. https://commons.wikimedia.org/w/index.php?curid=54927155

Some modern input devices have advanced in the way controllers can transmit action into response; one such feature is capacitive touch sensors. This feature has been implemented by devices such as the Valve Index or Meta Touch series, which can detect individual finger movement [29]. This allows users to have more natural hand movements like waving, gripping or pointing, without having to rely on dedicated buttons.

Figure 6 : HTC Vive Headset and Controllers in Use

Source: ESA, Wikimedia Commons, CC BY-SA 3.0 IGO. https://commons.wikimedia.org/w/index.php?curid=62840930

2.4.3 Haptic and Wearable Interfaces

The sense of touch can be experienced beyond the real world. Haptic and wearable devices are able to transfer this sense into virtual environments by physically stimulating the user's body when certain events take place. Besides handheld controllers that only convey vibrations or haptic feedback, wearable devices such as gloves, vests, or even full body suits can recreate some physical sensations using motors, actuators or air pressure and temperature systems. Some examples include, haptic gloves simulating resistance while grabbing on to a virtual object that is heavy or may be slipping away, vests or suits generating targeted vibrations or temperature changes to imitate impact or heat while falling on a rough surface or into water.

These devices work by receiving the digital signals from the simulation and converting them into a physical response that the body can detect, which creates a two way channel of connection that allows communication between the virtual environment and the user. The illusion may still break if the timing or intensity of the physical feedback is delayed, which is why synchronization must remain a priority. These systems are mainly used by research teams and for industrial use due to their high cost. Wearable haptics are starting to show promise in fields like medical training, rehabilitation and safety simulations where the advancement in realism that these haptics and vibrations provide can positively affect how users retain necessary skills [30].

2.4.4 Tracking and Motion Accessories

Some VR rigs aim to maximize immersion, and this requires additional equipment like trackers, treadmills and external base stations that give the user a larger range of movement which deepens the sense of being in the virtual space. Base stations, also known as Lighthouse sensors, emit infrared light (IR) to detect the exact position of the user, using the headset and controller's positions. This gives the simulation engine an accurate understanding of where the user is standing within the designated area, which direction they are facing and how they will move.

Another way to track the user accurately within an area is to use motion tracking sensors that attach to the body like at the arms, feet, or waist to capture full body motion. These trackers allow the system to have a more natural mapping of the user's movement into the virtual environment, which makes common actions like walking or crouching feel more lifelike.

Omnidirectional treadmills push the boundaries when it comes to movement inside the simulation by allowing users to physically walk or run in any direction while staying in place.

They are equipped with platforms that have low friction so that physical steps can be easily detected and converted directly into movement inside the simulation.

However, accessories of this kind require careful setup, calibration, and additional space. They also have high costs and require regular maintenance, which makes them unsuitable for people who use it casually, and as a result they are mainly found in research labs, industrial training scenarios or specialized simulation centers.

2.4.5 Audio and Communication Systems

Sound plays a major role in the way virtual environments are perceived by the brain. Many newer headsets have included 3D or spatial audio which lets users hear sounds as though they were coming from a specific direction and even adjust that sound to make it seem further or closer to the user.

The way spatial or 3D audio works is by processing the sounds through multiple channels that can alter the sound to simulate how the human brain analyses direction by adjusting timing, loudness and frequency. This method of generating an illusion of depth and location, is termed Binaural Rendering [31].

Some high end HMDs can include microphones that are built into the headset which can process voices and allow for real time communication between users inside the simulation, like in multiplayer scenarios. This makes interaction more natural with others inside a common virtual space.

2.5 Software and Development Frameworks

While hardware provides the physical means for the user to access virtual environments, software defines how these environments are designed and how they are to be experienced. All the frameworks, development toolkits, and most importantly, engines make up the foundation of any VR application.

The most commonly used game engines for creating and running VR environments are Unity and Unreal Engine [32]. They provide developers built in systems for rendering 3D graphics, logically handling physics, controlling shadows and lighting angles, and making sure the audio is synchronized. Interactions, object behaviors, and how the data from the hardware's tracking connects to a visual response in real time can all be scripted by developers using these engines.

They also manage the frame rate output and latency to ensure that the simulation runs without any delay.

Unity is the most preferred choice among developers and researchers as it is known for their flexibility and compatibility with multiple VR devices. On the other hand, Unreal Engine is mainly used when high quality visuals are required, such as in simulations for entertainment or architecture and industrial training. Both engines include tools that help in monitoring performance and how well optimized the rendering, lighting, and responses to user input are.

Beyond engines, SDKs (Software Development Kits) and APIs (Application Programming Interfaces) such as OpenXR and SteamVR provide tools that allow different HMDs, handheld controllers, and tracking systems to be integrated all into one environment [33]. This ensures that these applications do not need to be rebuilt from scratch just to be compatible with other VR devices. There are plugins like FMOD, which is used for audio, and PhysX, which is used for physics in simulations. These add-ons are pre-built systems that help developers when programming complex scenarios.

Developers need dedicated platforms to create, test and develop their virtual environments. Some of the most famous tools are Blender, which is used for designing 3D models that can be imported into engines and be made part of the environment, while software such as Visual Studio or Rider serves as the platform that lets developers program the logic of each interaction and how everything functions in the virtual world, using languages such as C# or C++ depending on which engine is being used.

When it comes to testing, there are debugging tools made specifically for VR, such as Unity Profiler and Unreal Insights, which are used to measure memory usage, latency, and frame rates, to help developers track any bottlenecks in performance. In large team projects or where more than one developer is working on a single project, version control systems like Git help the team track changes made by different members, merge updates and manage multiple branches made during development efficiently. These tools create a workflow that is not only organized and consistent but also safeguards against any errors or data loss by giving developers the option to revert to the last working condition.

3. FIRE SAFETY: PRINCIPLES AND PRACTICES

3.1 Fire Accidents and Hazards

Fires are one of the most common and destructive types of accidents to occur in both occupational and residential environments. While the origin of the fire can vary depending on the setting of the environment itself, majority of them originate from preventable causes [1], [2]. A fire breaks out when a sufficient amount of heat and a combustible material make contact with each other while having a steady supply of oxygen nearby [34]. In real life scenarios, these materials aren't intentionally placed in close proximity to each other, but often come in contact due to faulty systems or infrastructure, or through human negligence.

Electrical malfunction is a major category when it comes to fires and continues to be one of the leading causes of structural fire hazards [2]. A common occurrence is an overloaded socket powering multiple high wattage appliances, which can overheat over time and melt the insulation, eventually igniting dust or the socket itself. Another example of human negligence is ignoring regular electrical maintenance. Wires are coated in plastic or rubber casings to prevent the copper from touching other surfaces. Sometimes this coating wears off due to bending, rodents or from general wear and tear over time. Once the bare copper is exposed, it can touch another wire or metal surface and create a short circuit that releases sparks and intense heat, causing a fire. In principle, when electricity moves through an appliance, some energy is lost as heat, and this electrical resistance is normally harmless, but excessive current flow due to overloads or faulty connections can generate heat and cause significant damage. According to EU Fire Information Exchange Platform (FIEP) data, electrical causes account for roughly one-third of reported structural fires [4].

Another leading cause occurs in cooking environments such as homes, commercial kitchens, or school cafeterias. Most kitchens have oil fryers or other equipment that reach extremely high temperatures and if left unattended, the oil and fat in these can reach their flash point very quickly, once ignited they can spread flames across nearby surfaces within seconds. Portable heaters or space heaters placed too close to flammable fabrics like curtains, bed sheets or furniture can easily overheat and ignite. Most older models do not include an automatic shut off switch, which makes this an even greater danger in the cold months when they stay running for long periods of time [2]. These instances are particularly dangerous since they tend to occur in places where there may be a large number of people, increasing the chances of injury.

In many public spaces there are designated smoking zones or flame free policies which have been introduced because open flames and smoking materials remain a fire hazard. Open candles left unattended, especially inside the house, can easily ignite something flammable nearby. Incense burners may seem safe because there is no lit flame, but the glowing tip can reach temperatures up to 400-600°C and placing them near curtains or unstable holders can quickly make materials burst into flames. Even cigarettes can be dropped on a sofa or bed and burn through it slowly igniting the fabric, or be improperly discarded into trash bins with paper or other flammable materials in it [3].

Fires involving flammable liquids and gases often occur in industrial or laboratory environments where substances like LPG (liquefied petroleum gas), ethanol, or acetone are frequently used. Liquids that are classed as flammable don't just burn when they are lit on fire, but these liquids constantly emit invisible vapors into the air especially when left uncovered or heated and are highly flammable. These vapors don't need an open flame to ignite. A tiny spark from static electricity from synthetic clothing for example or a light switch is enough to ignite the entire cloud instantly. When containers aren't properly sealed or when valves have leaks , vapors can keep escaping and start filling the area. Without proper airflow in the room these vapors don't get ventilated and instead form a larger cloud. Unlike a solid fire which burns where the fuel is, vapors can travel several meters away from the source before igniting [34].

In more specialized environments, chemicals can react with oxygen or other substances that are mixed or stored together. They gradually keep heating up and when that pressure cannot escape, the temperature sealed inside eventually reaches the ignition point and bursts into flames without any spark or flame to ignite it. Linseed oil and other drying oils are often used to give wooden surfaces a polished and painted finish. When these oils are applied to rags and left crumpled in a pile or stuffed in a container, they start reacting with the oxygen in the air and this reaction releases heat. Over time, this temperature can rise until the cloth eventually catches on fire [35]. Finally, natural causes , although not common, can also lead to fires. Lightning strikes and volcanic activity can easily pose fire risks. Lightning that hits directly on the ground releases enormous electrical energy and can reach temperatures up to several thousand degrees Celsius [3]. Without proper lightning rods, these can ignite roofs of houses, causing severe destruction.

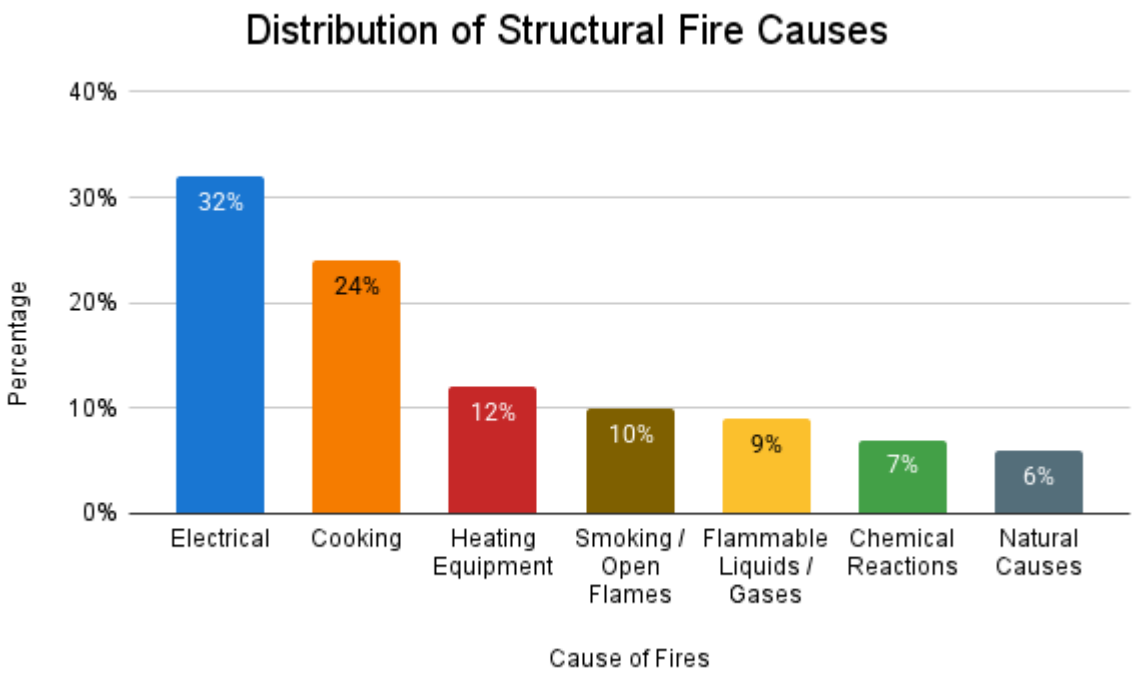

Figure 7 : Distribution of Structural Fire Causes.

Source: Data compiled from NFPA, Eurostat, EFSA reports [1], [2], [4].

In all these situations, the common pattern is the accumulation of heat near a fuel source in oxygen rich air. Maintaining such equipment before the chain of ignition ever begins and storing materials safely while keeping them monitored for heat sources is the most effective method of fire prevention [5].

Globally, the majority of reported fire incidents occur in residential environments, mainly because of human negligence and the presence of everyday combustible materials [1], [2]. Structural fires lead to thousands of injuries and significant property loss every year, which makes understanding how these incidents originate an important factor in recognizing fire as both a behavioural and physical hazard.

3.2 Classification and Behaviour of Fires

3.2.1 The Fire Triangle and Tetrahedron

Every type of fire, despite its size, depends on a combination of a few specific elements that make combustion possible. Commonly known as the Fire Triangle, it consists of fuel, heat, and oxygen, and when these elements are present in the right proportions, it causes a chemical

reaction and produces fires. If any one of these elements were to be removed or blocked in someway, it can cause the fire to weaken and eventually die out.

Ignition needs a source or initial energy which is provided by heat, this can be heat from friction, electrical energy, chemical reaction, or other sources capable of raising temperature high enough to reach the ignition point. Once a material combusts, the fire itself is able to generate enough heat to maintain the reaction. Any substance that can burn is classed as fuel, this can include solid materials such as paper and wood, liquids such as gasoline and alcohol, or even gases such as propane and butane, these are a few examples of the many sources that fuel fires. Oxygen plays a major role in supporting this chemical process by reacting with the fuel and allowing it to produce energy in the form of heat and light. In most scenarios, the air around us provides sufficient levels of oxygen to promote and sustain combustion.

The fire triangle effectively uses fuel, heat, and oxygen to explain the basics of ignition, but it does not take into account the chain reaction that keeps the fire alive once it actually starts. The fire tetrahedron solves this issue by adding the chemical chain reaction as a fourth element to represent this continuous cycle. This concept better describes how heat and oxygen interact with fuel to release more energy, which in turn creates new reactive particles that allow the fire to continue growing. A fire will burn as long as the chain reaction remains active, even if part of the fuel is removed.

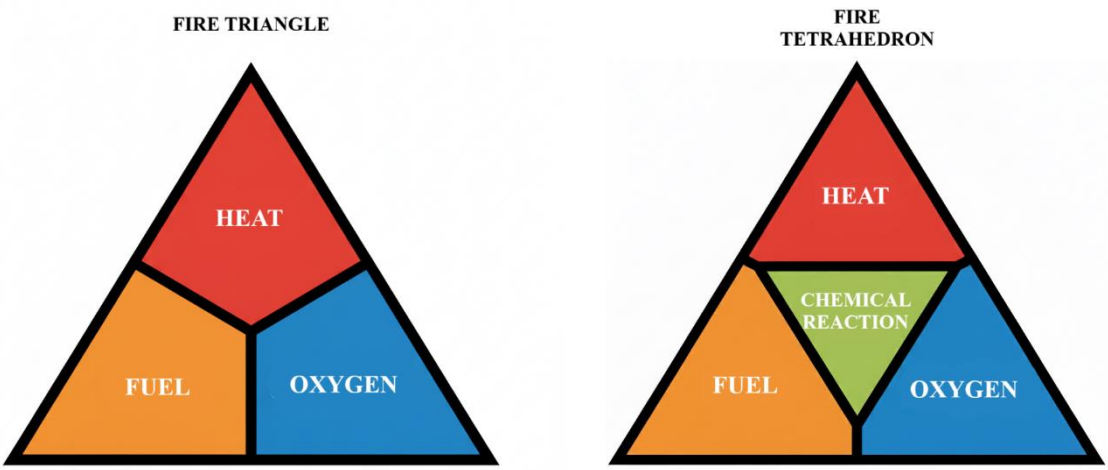

Figure 8 : Fire Triangle and Fire Tetrahedron Illustrating the elements required for combustion

Understanding the fire triangle and its tetrahedron form comprehensively is essential in order to suppress fires. Every firefighting technique works on one principle, which is to remove or disrupt one side of the triangle. Cooling with water removes the heat, smothering the fire eliminates its oxygen supply, and clearing out flammable materials gets rid of the fuel. Specific extinguisher agents such as dry powders target the chemical chain reaction directly by breaking the process down at a molecular level and stops the fire at its source. These principles form the core foundation in controlling fires and developing strategies to prevent them [5][6].

3.2.2 Classes of Fire (A–F)

Fires can be categorized into different classes depending on the type of fuel that is burning. Separating fires into these categories helps in identifying the type of fire and the best method to suppress it. It also provides a standard classification among all types of environments whether it is educational or industrial, which lets people who have even a small amount of fire safety knowledge quickly react using the right equipment. This plays an important role in safely choosing the right extinguisher with which to fight the fire, as using the wrong extinguisher can not only be ineffective but may also cause the fire to spread further or start dangerous reactions. Every class of fire stands for certain materials that burn differently. Some fires move quickly, some give off more heat than others, and some can ignite again right after they have been extinguished.

Class A fires involve solid combustible materials such as paper, cardboard, wood, fabrics and certain types of plastics. These materials are present almost everywhere in homes, schools, and offices, and cause the most frequent type of fire. Class A fires generally burn slowly and form embers that glow and continue to release heat even once the flame itself fades away. The key to extinguishing such fires and to make sure they do not reignite, can be achieved by cooling the material down until the temperature is below its ignition point. In this case, water based and foam based extinguishers are the most effective since they absorb heat efficiently and are also able to soak into the material to ensure the fire stays out.

Class B fires involve liquids that are flammable such as alcohol, oils, petrol, and paint. These liquids are more unpredictable than fires with solid fuels since they can evaporate quickly and release vapours that ignite even faster than the liquid itself. Unlike class A fires, these materials do not absorb water, they instead float on the surface. Using water to extinguish the fire in these scenarios can cause the flames to spread outward instead of extinguishing them. For this reason,

they are best controlled by cutting off the supply of oxygen by using foam, carbon dioxide or dry chemical extinguishers, which create a barrier between the air and the vapor to suffocate the flame.

Class C fires involve flammable gases like propane, butane and natural gas. This type of fuel can quickly escape and spread throughout the air and form invisible pockets of gas that can rapidly ignite, and if it's concentration in the air is high enough, it may even cause explosions. As long as the gas continues to leak, it keeps feeding the flame and cannot be controlled unless the supply of gas is cut off. For this reason, rather than attempting to extinguish the flame directly, the correct method to be followed is to isolate the gas source and remove oxygen.

Class D fires involve combustible metals such as sodium, magnesium, titanium, and potassium, which burn at very high temperatures. If these metals come into contact with water, foam, or certain chemical agents, they can react explosively, which makes them far more dangerous than regular fuels. When metals of this type are ignited, they produce extremely bright sparks and molten fragments which can cause the fire to spread to nearby surfaces. Due to the violent reactions that water can cause, Class D fires can only be smothered by using specialized dry powders that absorb heat from the burning metal and form a barrier to prevent the reaction from continuing.

Class E fires, often called electrical fires, occur in environments with energized electrical equipment such as circuit breakers, outlets and wiring or appliances [2]. Unlike the other classes of fire, electricity itself does not burn, but a vulnerability in the system can generate enough heat to ignite any material around it. Considering these environments, insulation, plastic or nearby fabrics can easily catch on fire. Another element that adds to the dangers of electrical fire is the risk of electric shock while attempting to extinguish a fire of this nature. Because these types of equipment have current continuously passing through them, attempts to physically access them can have extreme risks. For this reason, extinguishers require non-conductive agents such as carbon dioxide or dry chemical powders, so they can put out the fire without allowing the electric current to pass through the extinguisher or the person holding it.

Class F fires, also known as Class K fires in some systems, involve cooking oils and fats, mainly in large kitchen pans and deep oil fryers. These oils can heat up to very high temperatures and turn volatile. Even after the fire is put out, it has the ability to suddenly re-ignite and burst into flames. A common misconception is to add water, but since oil has a lesser density than water,

it floats and sits on the surface of the water. Adding water at this stage when the oil is at an extreme temperature can cause it to splatter violently and throw burning droplets in all directions, causing the fire to spread to other nearby materials. For this reason, wet chemical extinguishers are the most effective option as they form a thick soapy layer and cool the oil, effectively creating a barrier that prevents oxygen from reaching the surface and stopping the fire from re-igniting.

Classifying fires in this manner ensures that the right method of suppression is chosen for any situation, reducing the risk to people and property. When it comes to emergency response, understanding these categories is crucial to identifying hazards quickly and seamlessly applying the right measures to control the situation. This knowledge bridges the gap between recognizing the behaviour of fire and applying the correct procedures [5].

3.3 Fire Safety Measures and Protective Equipment

3.3.1 Detection and Alarm Systems

Fire alarm and detection systems are crucial in identifying a fire at its earliest stage before it becomes too big to control. The main function of these detectors is to sense abnormal levels of heat, smoke or flames and alert people in the building so that they can evacuate the premises quickly before the fire gets out of hand. These sensors play a major role in reducing injuries, minimizing property damage and assisting emergency responders by sending warnings as early as possible. Even a few seconds can make a huge difference since fires tend to grow rapidly once they reach the stage of spreading to their surroundings. These systems are also able to help pinpoint where the fire is located, allowing first responders to act more efficiently when entering the building.

One of the most common devices used is the smoke detector, and these are usually placed in rooms, hallways, and indoor spaces in general to sense if there is any smoke present in the air before the fire spreads any further. Modern smoke detectors use different sensing techniques. Some use ionization detection, which responds faster to quick moving flames, whereas photoelectric detection is more sensitive to slow fires such as burning fabrics or wiring. Most modern buildings use detectors with combined sensors so that all types of fires can be detected early, providing a more reliable level of protection. These types of dual sensing detectors also help in reducing false alarms, since they are able to distinguish between harmless smoke particles in the air and early signs of a real developing fire.

Figure 9 : Ceiling-mounted Smoke Detector Used for Early Fire Detection.

Source: By 최광모 – Own work, CC BY-SA 4.0. https://commons.wikimedia.org/w/index.php?curid=38364505

Another type of fire detection system is the heat detector. They are mainly used in environments where smoke may normally be present such as in kitchens, boiler or heater rooms, warehouses, and industrial production rooms. These detectors activate and send signals when one of two things happens, either the temperature inside the room reaches a certain threshold, or the temperature in the room rises unusually fast within a short period of time. Unlike conventional smoke detectors, they do not react to steam, dust or everyday cooking fumes. This makes them more reliable in environments where false alarms are common, such as commercial kitchens or industrial settings where the emission of smoke and vapor is a normal part of day to day activities. Despite the fact that they do not provide warnings as early as in the case of smoke detectors, they still play a very important role in detecting fires that produce little to almost no smoke in the beginning.

Figure 10 : Example of Heat Detector.

Source: By Georg Pik – Own work, CC0. https://commons.wikimedia.org/w/index.php?curid=25474741

Flame detector systems are a more advanced form of detection. These devices do not need the room to fill with smoke or even reach high temperatures before triggering. They react much faster than smoke or heat detectors when it comes to identifying fires by sensing the ultraviolet (UV) or infrared radiation (IR) that is naturally emitted by flames. These devices are mostly found in environments where fires can rapidly ignite and instantly grow out of control within seconds, such as laboratories or industrial spaces. Flame detectors are especially useful in areas where fuel is stored since even a small spark can become dangerous and life threatening very quickly. Their ability to respond fast and alert people gives workers an immediate warning in scenarios where every second counts.

Sprinklers are an important part of fire suppression systems. Each sprinkler head is equipped with a glass bulb filled with liquid that expands and breaks under heat, or a metal link that melts when exposed to high temperatures. Both types are very sensitive to heat, which is why each sprinkler head is activated individually unlike the common misconception that all sprinklers in the building activate at once. In reality, only the sprinklers closer to the fire will open, which allows the system to target the fire specifically while making sure the other parts of the building do not get damaged by water unnecessarily. This type of system reacts very quickly during early

stages of a fire, before the flames have spread far. This makes them highly effective in containing the fire to one room or floor. Due to this, many large buildings have safety regulations that require them by law to install sprinklers as a part of their automatic fire control and response systems.

Figure 11 : Automatic Fire Sprinkler with Heat-sensitive Glass Bulb.

Source: By Brandon Leon – Flickr: Day 25: Fire Sprinkler, CC BY-SA 2.0.
https://commons.wikimedia.org/w/index.php?curid=28858271

Alongside all these systems, manual alarms also play an important role in fire safety. Break glass points or pull stations allow people to manually activate the building's fire alarm system when they see danger. Manual activation of this kind is especially important in cases where the conventional automatic detectors have not yet detected the fire or triggered the alarm, giving occupants a way to raise the alarm the moment they see something unusual. Once it is triggered, the alarm alerts everyone inside the building and sends a signal to the control panel that handles fire alarms. This panel acts as the centre of the entire system, showing where the fire was detected, which helps first responders or firefighters understand the situation and pinpoint exactly where the incident was spotted, helping them plan the best course of action.

These detection and alarm systems together form the first line of defence against the serious damage and injury that fire can cause. Their main function is not to extinguish the fire, but to detect it as early as possible and give people enough time to escape safely and allow first responders to act efficiently. [5][6]

3.3.2 Fire Extinguishers and Usage Principles

Fire extinguishers are some of the most efficient tools for suppressing fires before they spread out of control. Most extinguishers are stored under pressure and are equipped with a handle mechanism that releases the substance when squeezed. They work by releasing a pressurized agent such as water, foam, carbon dioxide, or chemical powders through a nozzle. These agents aim to eliminate heat, cut off oxygen or interrupt the fire's chemical reaction.

Extinguishers are designed for different types of fire scenarios:

- Water extinguishers are mainly used for Class A fires that involve burning solid materials such as paper, wood, and fabrics. These materials are soaked by the water to cool them down and stop the fire from spreading.
- Foam extinguishers are mainly used for Class B fires that involve flammable liquids. The foam forms a blanket-like surface over the liquid and blocks its oxygen flow, stopping the vapours from being able to ignite again.
- Dry chemical extinguishers can be used for Class A, B, C, and E fires because the powder agent that is used, quickly interrupts the chemical reaction and puts out the flames almost instantly. This also makes them very efficient for electrical hazards.
- Carbon dioxide (CO2) extinguishers can be used for Class B and E fires since CO2 displaces the concentration of oxygen around the flame and does not leave any residue. This makes it especially safe for sensitive electrical equipment and devices.
- Wet chemical extinguishers are specifically made for Class F fires that involve cooking oils and fats that are used in pans and deep oil fryers. The chemical forms a cold soapy layer directly on top of the hot oil, preventing it from reigniting.

To operate an extinguisher safely, there is a simple method known as the PASS technique. PASS stands for Pull, Aim, Squeeze, and Sweep.

- Pull refers to removing the safety pin, which is located near the handle. This unlocks the extinguisher from its locked safety position.
- Aim refers to pointing the nozzle of the extinguisher at the base of the fire rather than at the flames itself.
- Squeeze refers to squeezing the handle once the nozzle is aimed correctly, which then releases the extinguishing agent.

- Sweep refers to the sweeping motion the user must perform so the nozzle moves from side to side, this helps evenly cover the entire burning area.

Following this method ensures that the fire is being targeted directly at its source and also reduces the risk of missing areas where the fire could reignite. PASS is very commonly taught in educational and workplace safety training because it provides a clear and quick way for any person to use an extinguisher under pressure just by remembering one simple acronym.

Even when extinguishers are used properly, they still have limitations that need to be considered carefully. Extinguishers are mainly designed for smaller fires that are still in the early stages after ignition because after the flames grow too big or start spreading quickly, the extinguisher may not be powerful enough to stop it. While the user is attempting to put out a fire, they must keep a safe distance and make sure there is a clear path behind them in case attempts to extinguish are futile and they need to escape. The situation can instantly become dangerous if the smoke becomes too thick or if the heat rises suddenly, this would be the moment to immediately stop and evacuate. Understanding the strengths are weaknesses of the extinguisher makes it safer to use efficiently and helps prevent small accidents from turning into something far more serious. [5]

3.3.3 Personal Protective Equipment and Suppression Tools

Personal protective equipment (PPE) and basic suppression tools are important parts of fire safety. While automatic systems such as detectors, alarms, and sprinklers help in detecting and controlling fires early, the function of PPE is to protect the people who need to respond to dangerous situations quickly before the first responders arrive. These tools do not replace the heavy-duty gear that firefighters use, but they give users a safer way to manage small incidents before they worsen. They also help the user in guiding others during evacuations while staying protected until they are able to escape.

One of the most common suppression tools is the fire blanket. Fire blankets work by cutting off the fire's oxygen supply. They are able to do this because they are made of heat resistant materials such as fiberglass and are designed to smother smaller flames. These can be especially effective if a person's clothes accidentally catches on fire. Its simple design makes simple for people with no training since all they would need to do is place or wrap the blanket over the flames to starve the fire.

Figure 12 : Wall-mounted Fire Blanket for Smothering Small Flames.

Source: By 247homerescue – Own work, CC0.https://commons.wikimedia.org/w/index.php?curid=33348184

Basic PPE also includes heat resistant gloves, which allows individuals to touch hot objects like open doors with heated handles or remove smouldering materials safely without burning themselves. Since smoke spreads much faster than fire, having a small layer of respiratory protection can make a big difference in navigating through hallways safely without risking disorientation. Fire resistant masks or smoke hoods are able to provide a decent amount of protection by filtering smoke and giving the user a few more minutes of breathable air during intense evacuations.

Most workplaces and schools also keep hose reels, and sometimes even fire buckets, which are filled with sand or water. This is an older practice but still effective in suppressing small flames in specific situations like Class A fires. Sand buckets are especially useful when electrical systems or liquids are present since sand can smother flames without risking any chance of getting an electric shock or spreading oil fires.

In hazardous environments like laboratories, industrial workshops and chemical storage rooms, less common PPE can be found. Gear such as face shields, aprons and protective eyewear are all designed to block sparks, splashes of dangerous materials or chemical vapours. This type of gear helps in preventing injuries if an emergency shutdown takes place, or if accidental

reactions occur, giving the workers enough protection to move away as quickly as possible from the hazard without being seriously injured.

Professional firefighters rely on much more advanced types of PPE. This includes heat resistant suits, reinforced helmets and boots, and breathing apparatuses that allow them to enter high heat situations and stay in smoke filled environments for longer periods of time. This shows the huge contrast between the PPE that is available to the public and the highly specialized gear used by trained professionals, further highlighting why untrained individuals should never attempt to fight large or fast spreading fires even with the PPE they may have at hand. [5][6]

3.4 Fire Prevention Policies and Evacuation Procedures

3.4.1 Prevention Policies and Education

Policies on fire prevention are very important as these safety strategies come into play long before any actual fire breaks out. Their purpose is to reduce the chances of ignition and minimize the presence of unstable fuel sources along with making sure that individuals understand how to safely respond in the case of a real fire. Most incidents start from simple issues such as poor maintenance, unsafe behaviour, or lack of awareness. These policies focus on reducing hazards and teaching people how to accurately recognize warning signs, this ensures that the general public knows what to do before a situation becomes dangerous or life threatening. Emergency equipment is only as effective as intended when the user is well informed on how to use it efficiently because fires can spread quickly and even a few seconds of hesitation can make the situation worse. [5][6]

Prevention begins with education. Schools and universities often provide basic safety instructions during orientation seminars and teachers instruct students on how to behave around electrical equipment, cooking appliances and chemicals. Some of these simple precautions can also be followed at home, habits like unplugging unused devices, making sure the kitchen is not left unattended while cooking, storing flammable liquids properly, and much more. Keeping these basic rules in mind can significantly lower the risk of fire. Workplaces and some universities usually take an extra step by including mandatory safety lectures or online training sessions that must be completed regularly. These programs help people better understand how fires start and what their responsibilities are when such an event occurs. [5]

Maintenance is one of the main aspects of prevention policies. Many fire incidents happen because old equipment haven’t been inspected often enough or maintained well, such as

electrical wiring getting frayed, overloaded sockets, old space heaters, and faulty appliances. These common sources of ignition are the reason buildings are required to have regular scheduled inspections by trained technicians to check the condition of devices like alarms, extinguishers, sprinklers and ventilation systems. Most public buildings and workplaces keep maintenance logs so that they can keep track of any worn out or malfunctioning equipment that may need replacement before it has the chance to become a hazard. Conducting simple routine checks like these in the house can also be very advantageous. Things like cleaning the stovetop, replacing damaged cables and keeping the fuse box in good condition can reduce risk by a lot. [5]

Having control over the environment is another important part of prevention. Fuel sources are one of the most dangerous factors and accelerants of fire. These fuels need to be kept away from potential ignition points, which can be done by storing flammable liquids in proper containers, keeping paper, wood or fabrics far away from heaters, regularly cleaning grease build-ups in kitchens and workshops. Workplaces that normally deal with hazardous materials already have stricter rules. Making sure chemicals are stored in specific storage cabinets, spill-proof containers, and correctly naming them using labels so that people know exactly what materials they are handling, are all important policies that help create a safer environment not just for the trained personnel but also for everyone else who uses the building. [6]

Many people underestimate fires and assume that small fires are easy to control. For this reason, prevention policies encourage frequent reminders in the form of posters or instructions on walls, along with clear signs showing the location of extinguishers, emergency kits, alarms, and directions to assembly points. In the everyday home, it can be as simple as placing a fire safety board in the kitchen and teaching family members how to react if something does catch fire. In larger buildings, the staff usually receive short demonstrations on how to use extinguishers or how to identify if something may overheat and start a fire or recognize burning smells which can alert people of danger without actually seeing the flames. [5]

Lastly, prevention also relies on encouraging responsible behaviour. Simple rules like not blocking escape routes, prohibiting storage in stairwells, making sure emergency doors remain unlocked during working hours when the public is present, encouraging people to report anything unusual like burning smells, overheated appliances, flickering lights, sparks from sockets or damaged wiring, because even if it turns out to be a false alarm, it is safer than

allowing these issues to escalate into something dangerous. When everyone accepts and follows these behaviours, the risk of fire reduces dramatically. These policies and education together create proactive measures to enforce fire safety, as a well-maintained environment supported by consistent training and proper habits create a safer space for students, employees and families. [6]

3.4.2 Evacuation Procedures and Drills

When the fire alarm is pulled, the first priority is to get everybody out of the building as quickly as possible in an organized manner. Fires spread far faster than most people expect. Evacuation procedures exist because even a few seconds of hesitation can make a small incident become something dangerous and unmanageable. People are instructed to immediately stop what they are doing and move towards the nearest exit without trying to grab any belongings, no matter how important, or even trying to investigate the alarm itself. Even if it eventually turns out to be a false alarm, reacting instantly to the alarm is always the safest response because the earlier the evacuation, the more injuries that can be prevented. [5][6]

Once the evacuation procedure begins, the manner in which people move through the building is just as important as how quickly they react. Elevators can easily shut down or malfunction during a fire and expose people directly to smoke. Because of this, staircases are always the safest option. Everyone should move calmly, avoid running, and keep to one side of the staircase, usually the right side, so that other people behind them can pass if needed. In the hallways especially, smoke can rise very quickly and form in large amounts. Staying low to the ground in these situations helps in breathing easier and also makes it easier to see ahead. As people make their way away from the fire, it is a good practice to close the doors behind them, without locking them, so that the spread of smoke and flames is slowed down, giving the others a safer path to escape. Universities, malls and office buildings often depend on emergency lights to guide people along long corridors and stairwells in case the electricity completely goes out. These emergency strip lights allow people to follow the evacuation route even when a power outage and smoke reduce their visibility. [5]

Figure 13 : Standard ISO 7010 Emergency Exit Symbol.

Source: By MaxxL – Public Domain. https://commons.wikimedia.org/w/index.php?curid=25959039

In some larger buildings, there are trained fire marshals who help guide people toward the safe exits and also check rooms to make sure there is no one being left behind. Their responsibilities are to assist individuals who might hesitate or panic, make sure all the escape routes are clear and there are no blockages, and most importantly communicate directly with the firefighters and first responders about information like which floors were checked and cleared, where the fire was first noticed, and if there might still be someone stuck inside. Having staff members designated with such responsibilities keeps the evacuation process organized and reduces panic and worry. [6]

Schools follow similar practices but in a much more structured manner. When drills are conducted, or real emergencies take place, teachers lead their classes out of the building while making sure everyone stays together and that younger students who may be more prone to confusion and panic stay close. Many institutions also assign student helpers or class marshals who assist with simple tasks such as closing the classroom door, helping the teacher guide classmates down the hallway, or supporting anyone who may feel overwhelmed by the whole situation so they can evacuate as soon as possible. These roles may seem small, but they make

evacuation and emergency routines much smoother and help teachers take better control of the situation for their class. This organised structure of responsibility makes it easier for the staff to ensure that no one is left behind, especially in educational environments where there are typically a very large number of students in a single building. [6]

After evacuating the building, everyone must make their way to the designated assembly point, which should be an open and safe area far away from the structure so that any further damage that may happen to the building doesn't put anyone at risk. In schools, teachers then take roll calls to confirm that all students are accounted for. Only once the firefighters arrive and give the green light can people be allowed to leave the assembly point or re-enter the building. Even if the fire looks small from the outside, the interior may still contain smoke pockets, or hidden hotspots that can reignite. The fire could have also weakened and damaged a lot of the structure causing it to collapse. Staying together in one location allows first responders to work more efficiently until they have analysed the entire situation and dealt with the dangers, preventing any unnecessary risk and confusion. [5]

Sometimes these procedures include people who may need additional assistance. People with limited mobility may require gear like evacuation chairs or assistance while navigating through hallways and staircases, many buildings train specific staff members to handle situations like this safely. Some facilities have lift systems that are specially designed and approved for use during fire evacuations, but these are only meant to be used in controlled conditions. Trained personnel are taught to guide certain individuals calmly and ensure they continue moving towards the exit, because panic, anxiety and disorientation are very common reactions during emergencies. Scheduling regular evacuation drills reinforce these procedures by giving everyone the chance to practice them in a predictable and controlled environment, and their repetitive nature helps build familiarity which increases confidence so that occupants can respond quickly during real emergencies. [6]

4. VR AND EDUCATION IN FIRE SAFETY TRAINING

4.1 Introduction to Fire Safety Education Research

Fire safety education has always played an essential role in reducing injuries and preventing property loss. But the way people learn about fire hazards has changed over time. Earlier types of training mainly focused on following instructions and rules, usually through classroom lessons or printed instructions that explained to people what they should do when an alarm does goes off. Eventually, researchers realised that just hearing or reading information is not enough during a real emergency, because people react very differently when they are under stress. This changed how people viewed fire safety and research toward understanding not only what people learn, but how they behave when they are suddenly placed in a dangerous and fast-moving situation, became more relevant. A lot of modern fire safety research now tries to focus on human behaviour during emergencies. Studies show that when people face unexpected alarms, their reactions depend on factors such as stress, previous experiences of the same type, and how confident they feel about the steps they need to follow in such scenarios. Even when individuals know the correct theory and procedure in their mind, having to deal with loud alarms, limited visibility, or crowded hallways easily cause hesitation and confusion. This has led researchers to look more closely into the psychological side of safety training, especially how attention, memory and decision making skills can quickly change in situations where pressure is high and the time is limited. [5][6]

Another idea that keeps coming up is that learning is more reliable when the training environment feels realistic, or familiar and comfortable. Many studies highlight the difference between understanding instructions in a calm setting compared to actually having to apply those same instructions during a real emergency. People may know their nearest exit route, but when the alarm scares them, they might still follow the crowd or take longer paths because their attention changes under stress. In other words, knowledge on its own is never enough. The way information is practised plays a major role in whether someone can recall it quickly when they need it most. [6]

After these findings, researchers have explored different ways to improve how fire safety education can be given. Some highlight the importance of repetition, since people tend to forget the details of evacuation procedures after not having any practice in long periods. Others point out that training can become more meaningful when the learners experience scenarios that mimic real-life conditions, even if those conditions cannot be reproduced perfectly. Ideas from behavioural science and repeated exposure together form the basis of many modern discussions about what effective safety training should look like. [5][6]

4.2 Traditional Fire Safety Training Methods

4.2.1 Common Training Approaches

Most schools, workplaces, and public institutions depend on a few standard methods when it comes to fire safety training. These approaches are very commonly used because they are simple to organise and can work on large groups of people without needing special equipment or advanced planning. The most common one is the traditional fire drill. Drills usually have a predictable routine, when an alarm is sounded, everyone is asked to leave the building through

the nearest exit while staff members guide the movement. The main aim of this is to help people become familiar with evacuation routes and understand how they should behave when an alarm goes off [5].

Short instructional sessions are another common method, like an orientation program for new students or employees, they are usually conducted occasionally throughout the year. In these sessions, instructors explain the basics about where the exits are, how the alarm system works, and what people should avoid doing during an emergency. These lessons are usually short, so they focus on the essential parts without going into detail. They work more like reminders than a training. Some institutions also do live demonstrations, especially when they want to show people how fire extinguishers or alarms work. A trained safety officer explains how to operate the equipment and may show an example of extinguishing a small flame. Seeing this process step by step can still make the actions easier to remember rather than if they were only described in words [5].

Alongside this, many places include basic safety instructions in their onboarding process. New employees might receive a short briefing from their supervisors, and students are introduced to simple rules by teachers at the beginning of the year. These explanations usually cover the same core topics, same as drills and demonstrations, just in a more compressed way, so that people at least know the basics of how to respond without being overloaded with information on the first day. Taken together, these approaches form the backbone of fire safety training in most environments and are the default option in many institutions only because they are easy to repeat and relatively cost lesser [5].

4.2.2 Effectiveness and Limitations in Real Emergencies

Traditional fire safety training does sometimes help people understand the basic steps they should follow when an alarm goes off, and most studies agree that drills do improve general awareness [5]. After a drill, participants usually remember the nearest exits, they understand the flow of movement, and they feel more familiar with the layout of the building. This familiarity reduces the initial shock that many people experience when they hear an unexpected alarm, giving them a rough idea of what to do next.

The problem is that this effect does not always last, as research shows that memory of evacuation routes fades quickly, especially when drills are only done once or twice a year. Many people remember the steps right after training, but a few months later they struggle to remember that same information with confidence. Surveys in schools and universities also point out that participants often treat drills as routine tasks rather than real learning moments, which means they pay less attention to the details that actually matter during an emergency [9].

Real fires also involve conditions that drills cannot safely match. Loud alarms, heat, smoke, poor visibility, and people moving in different directions all create stress, and stress changes the way decisions are made [6]. Most drills take place in calm, controlled situations, so participants rarely experience the confusion that appears when everything goes wrong all at once. Studies mention that hesitation, freezing, and blindly following the crowd are common behaviours in real emergencies, even among people who have completed training before [8].

Organizing these events have their own issues too. Drills in large buildings can be difficult to plan, especially when there are hundreds of people involved like with younger students or anxious individuals who may panic even during a controlled drill, which forces instructors to keep the exercise gentle. Workplaces sometimes rush through the process just to avoid interrupting company operations for too long. So the training becomes more of a formality than a meaningful experience. As a result, participants may remember the basic instructions but still lack the confidence needed to act quickly when something actually goes wrong [9].

Institutions keep using these methods because they are cheap and familiar. They do provide a useful starting point, but it is clear that they do not fully prepare people for the pressure and uncertainty of real fires, which is why a lot of new attention has shifted towards more immersive training techniques [8].

4.3 Virtual Reality as a Learning Tool

4.3.1 VR in Education and Skill Development

Virtual Reality has changed the way people learn by making the experience from something passive to something more hands-on. Instead of just listening to an explanation or watching a demonstration, learners can enter a digital environment and try things themselves. This kind of participation makes a big difference in subjects where understanding comes from actually doing the task rather than memorising its steps. VR gives people a chance to explore, interact, and solve problems inside a space that feels more realistic than an everyday classroom.

Medical education is one of the clearest examples of this. Students can practise surgical procedures or emergency responses in a controlled environment without risking any harm to real patients. Engineering students can look inside machines and test their mechanisms while working with systems that would normally be too expensive or dangerous. Science learners use VR to run experiments and visualise complex reactions that are difficult to demonstrate during a normal lab session. Fields like aviation, industrial safety, and emergency response also rely on VR to simulate high-risk situations that trainees might never be allowed to experience fully in real life.

Across these different areas, the results are that people who learn through VR often stay more engaged because they are involved in the process instead of just listening. This interaction helps them remember information better and understand it better. When learners are able to repeat tasks and get immediate feedback inside the virtual environment, it strengthens their memory and helps them build confidence faster. Many skills require more than just reading instructions. They require people to react, adapt, and make these decisions while something is happening in front of them. This is also why VR is especially relevant for training related to safety and emergency situations, where hesitation can be dangerous. [8], [11], [12], [13], [14]

4.3.2 Strengths and Challenges of Immersive Learning

Immersive learning environments such as VR work well because they place the learner inside a situation instead of making them watch from the outside. When the user's field of view is the virtual scene, they will have fewer distractions, and their mind treats the experience like there is something happening directly in front of them. This level of focus helps people process

information more clearly and remember it for longer. When it comes to skills that rely on physical actions, this engagement is still useful because the learner is involved in every part of the process. On top of that, VR allows people to make mistakes safely. In a real environment, some tasks can be dangerous, expensive, or difficult to repeat by someone with lesser experience. Virtual environments remove that concern. Learners can try different approaches, test their decision making, and repeat the same task multiple times without risks.

On the other hand, VR training does have some limitations. If the virtual environment does not feel convincing enough, because the graphics are low quality or because of awkward interaction, users may treat the scenario as a game rather than a serious situation. VR also cannot fully recreate details such as heat, smell, or the full mechanics of real-world movement without expensive advanced gear. These missing elements can reduce realism, especially in safety training where the environment plays a major role. Cost also add another layer because institutions need hardware and someone who knows how to maintain the system, which not every school or workplace will have. Some users experience discomfort such as motion sickness or eye strain, especially during longer sessions, so designers are sometimes forced to shorten or simplify some scenarios.

These issues do not make VR useless, but they do show that immersive learning needs thoughtful design to be effective. When the environment feels natural enough and the tasks match what the learner is supposed to practise, VR can become a strong tool for building skills that require clear thinking and confidence under pressure. [8], [11], [12], [24], [27]

4.4 VR for Safety and Emergency Response

4.4.1 VR Applications in Hazard Recognition

One of the most practical uses of VR in safety training is helping people learn how to recognise hazards before they become dangerous [11]. In real situations, one of the biggest challenge is not reacting to the fire itself but noticing the early warning signs that something is wrong. VR is useful here because it can place learners in environments that contain subtle hints of risks. Things that they might overlook in real life, and let them practise identifying these without any real consequences [8].

Studies done on workplace safety often highlight this point that when people are asked to walk through a digital version of a factory floor or an office environment, they are more likely to notice issues such as blocked exits, exposed wiring, or malfunctioning equipment because that scenario draws their entire focus to the environment as a whole [8]. In normal training sessions, these hazards are usually described through photos or slides. VR, on the other hand, presents the environment in a more natural way, so that learners notice problems by exploring the space rather than memorising a list of examples on a sheet [11]. It can also simulate hazards that would be too risky to demonstrate physically, such as early signs of smoke, overheating devices, or leaking gases, using visual and audio to teach what to look for.

VR systems allow trainees to inspect equipment, check electrical panels, or test alarms to see how they respond. This interaction helps learners understand how systems can behave differently and what maintenance issues might lead to a fire. It also encourages them to pay attention to details they might ignore in everyday life. When users repeat these tasks in different

scenarios, they start to develop a habit of scanning their surroundings more carefully, which strengthens their connection between observation and action and reduces the chances of a small issue turning into an emergency [12].

4.4.2 VR for Evacuation and Decision-Making

Evacuation is a difficult part of emergency response because people rarely get to practice it in a situation that feels serious. VR helps with this by placing users inside environments where they must react to alarms, navigate through hallways, and make quick decisions under pressure [11]. Users can hear loud noises, see flashing lights, or notice smoke in some parts of the room. These elements affect how people think, and VR lets them experience these challenges without being placed in real danger. It becomes clear very quickly that staying calm and following the correct route is harder than it sounds when the environment feels unstable [8].

Many people know their exit route only in theory, and under pressure they may follow the crowd or take longer paths without realising it. VR scenarios can test this by placing users in different positions in a virtual building and watching how they move once an alarm starts [11]. They have to choose between multiple halls, pay attention to signs, and avoid blocked paths. All of which mirrors the decisions they would face in an actual emergency. Some simulations also add simple moral or practical decisions, such as whether to help someone nearby or continue moving, which reflects the real behaviours that usually delay evacuations [6]. By repeating these scenarios, people get a better sense of which actions are safe and which responses increase risk, and trainers gain a clearer insight into who hesitates, who rushes, and who freezes when pressure appears.

4.4.3 Limitations in Safety-Focused VR Research

Even though VR has made safety training more interactive and realistic, the research also points out many limitations that effect how well these systems prepare people for real emergencies [8]. One of the most common issues is that VR cannot fully recreate the sensory complexity of an actual fire. Elements like heat, strong smells, or the way smoke behaves naturally are difficult to simulate accurately. Many VR systems rely mainly on visual and audio cues, and while those cues do help in creating a sense of urgency, they still leave out the important parts of the experience that influence human behaviour in real situations [11]. Because of this, some users may not react the same way in VR as they would during an actual emergency where sensory overload is much higher, which makes it harder to predict real-world behaviour based on virtual performance.

At the same time, practical challenges related to the equipment cannot be ignored. VR headsets can be expensive, and can require technical setup and need regular maintenance. Some institutions may not have enough devices for larger groups, leading to shorter or rushed sessions. In addition, not everyone reacts comfortably to VR. Motion sickness, dizziness, and eye strain are mentioned frequently [27].

4.5 VR Fire Safety Training Systems

4.5.1 Existing VR Fire Training Approaches

Research on VR fire safety has grown quickly over the past few years, mainly because traditional drills cannot recreate the pressure and sensory difficulty of a real fire. Many of the existing VR systems follow the same general idea. They place the user inside a virtual room, start a fire hazard, and ask them to respond using basic procedures such as identifying the fire, choosing the correct extinguisher, or evacuating to a safe area. Even though each system has its own design style, the goal is always the same.

A common approach seen in several studies is a single-room fire scenario. In this setup, the user is placed inside a small virtual space such as a kitchen, office, or workshop. A fire breaks out in one corner, and the user must decide whether to extinguish it or leave the room. This kind of scenario is popular because it teaches the early stages of fire response, especially in recognising the type of fire, selecting the correct extinguisher, and aiming at the base of the flame. Most of these training scenarios rely on basic interaction mechanics that let the user pick up the extinguisher, pull the pin, and operate it in a way that resembles real life.

Other VR training applications focus more on evacuation than extinguishing a flame. These simulations put users in larger virtual buildings such as school hallways or office floors with multiple exit routes. The exercises test how well participants identify the right path, avoid blocked exits, and follow emergency policies while dealing with alarms or smoke [12]. More advanced systems combine both aspects and use scenarios that change based on the user's actions. Ignoring early warning signs may cause the fire to grow and visibility to reduce, while quick responses keep the situation at hand. Some of these systems also track mistakes and provide feedback later on so learners can see exactly where they hesitated or didn't choose the safest option.

There are also VR tools that simulate industrial and laboratory environments. These are usually more technical and designed for workers who handle flammable chemicals, electrical equipment, or machinery. The scenarios teach identifying hazards, emergency shutdown procedures, and safe handling of equipment during fires. Because these workplaces involve more specialised risks than a typical home or office, the VR systems need to recreate specific equipment and layouts accurately so users can practise the exact steps required in their job. [8][11]

4.5.2 Behaviour and Learning Outcomes in VR Fire Scenarios

When users are placed in a virtual room where smoke starts building up or an alarm suddenly goes off, they tend to react with more urgency, and their decisions become more realistic than in a calm, scheduled evacuation [8]. This suggests that VR can trigger some of the same psychological responses that occur during real emergencies, which helps researchers study student behaviour more closely and gives trainers a clearer picture of how people actually respond when they feel a small amount of pressure.

One pattern that appears frequently is how quickly people try to locate the source of danger in scenarios where small flames appear or detectors start beeping, participants naturally look

around, assess the situation, and decide whether to extinguish the fire or evacuate [12]. Extinguisher training inside VR also reveals common mistakes such as aiming too high away from the base, standing too close to the flames, or forgetting to pull the safety pin. Because these errors are visible and easy to repeat, users can correct them over several attempts until the correct steps feel more natural [11]. Studies generally report that people are more confident in recalling evacuation routes after VR-based training compared to traditional methods, even if some learners still need time to get used to the controls of the scenario [8].

4.5.3 Realism Issues and Remaining Gaps

Even though VR fire training has become more advanced over the past few years, several studies still point out gaps that limit how realistic the experience can be [8]. One of the biggest issues is the difficulty of recreating fire exactly how it behaves in real environments. Flames in VR usually follow a fixed animation or scripted pattern, which means they do not spread with the same speed as an actual fire neither are they as unpredictable [12]. In real situations, factors like airflow, the type of fuel, and the materials present around the room all somehow affect how quickly a fire grows in different ways. VR systems rarely account for these variables, so the user may not get a full sense of how quickly conditions can worsen once ignition begins.

Smoke introduces similar problems where many simulations rely on particle effects that give a visual of smoke, but they do not reproduce how smoke physically restricts visibility and most importantly irritates breathing and causes disorientation [11]. Since real smoke moves unpredictably and can fill a room unevenly, the fixed smoke patterns in VR can feel less threatening than they should. While headsets can play alarm sounds or show bright flames, they cannot simulate rising temperatures or the discomfort that comes from being close to a fire, which are all factors that influence the decisions made in real emergencies.

Interaction with the environment also has its limitations. Extinguishers in VR often behave the same way regardless of distance, movement, or its angle. Users do not have to think about the recoil, weight, or the duration the extinguisher can spray for, which are important considerations in real life [8]. Building layouts are sometimes simplified, and crowd logic is difficult to simulate accurately. Real evacuations involve people blocking exits, slowing down corridors, and creating unexpected blockages near stairs. If these factors are missing, users may move through the virtual building more easily than they ever could in reality.

Finally, even though VR helps with learning and confidence, the research shows that not all skills fully transfer back to the real world. Some users report that they feel prepared in VR but still hesitate in physical drills, especially when handling real equipment [11].

4.6 Summary

The studies reviewed in this chapter show a clear pattern about how fire safety is taught and how people respond to emergencies. Traditional methods such as drills, demonstrations, and short instructional sessions are still the most common approach because they are simple to organise and easy to repeat. They help people get familiar with evacuation routes and basic procedures, but a lot of aspects becomes less effective when real pressure is involved. Many participants struggle with hesitating, losing memory about simple procedures, and difficulty in following instructions during unpredictable situations.

Research on immersive learning explains why more interactive methods are being explored nowadays. VR captures attention more naturally and encourages users to interact with their surroundings instead of relying on written instructions or controlled fire drills. When people are placed inside a scenario where the environment requires them to focus, they process information differently and build a stronger memory about it. Studies from education and high-risk industries show that learners tend to stay engaged longer, recall steps more accurately, and respond with more confidence after practising inside a virtual environment.

Taken together, this suggests that VR has a very strong potential for improving fire safety education, especially in areas where practical experience is hard to provide, but it works best as part of a wider safety strategy instead of a complete replacement for the existing conventional methods. It offers a controlled space to practise skills that matter.

5. SYSTEM DESIGN AND IMPLEMENTATION

5.1 Overview of the VR Fire Safety Simulation

The virtual reality simulation that has been developed in this thesis is designed to represent fire emergency scenarios where the user can experience the basic steps of responding to a fire. This project focuses on two indoor settings, an office and a kitchen, because these tend to be common places where fires usually start. They also represent different types of fire hazards. The user begins the simulation standing inside the room while the fire is already burning along with a fire crackling sound playing in the background to make the situation feel more realistic and to signal that something is wrong. After the user spawns in the room, they are free to walk and look around, and inspect the fire from different angles so they can decide how to react.

Figure 14 : Overview of Office VR Environment

Although the user can walk around and interact with the other objects in any order they prefer, the first thing the user should interact with is the fire alarm button. This button is placed in the room on the wall. When pressed or interacted with, the fire alarm sound starts playing, this simulates how real buildings warn people when a fire is detected by and the alarm is pulled. This is an important part of the experience because it mirrors the basic principle of raising the fire alarm, which is one of the first steps people should be taking during real fire emergencies. Once the alarm is activated, the user should make their way towards the wall where the fire extinguishers are mounted on hooks. This set of extinguishers includes one of each, for different

fire types, such as water, carbon dioxide, foam, and dry powder, and they are placed besides each other so that the user can think about which one is appropriate for the type of fire they are currently dealing with.

Figure 15 : Overview of Kitchen VR Environment

Once the fire alarm is pulled, the user needs to decide which extinguisher to choose, if an extinguisher is not interacted with within a few seconds, the correct one will be highlighted, this helps guide the user in case they cannot decide. This also reinforces the policy that acting fast can keep the incident under control. Once the right extinguisher is picked up, three short description cards appear on the wall above the extinguishers. These cards explain why the one extinguisher is the correct choice and also why the other two choices would be incorrect or even unsafe for that specific scenario. This gives the user a short minute when they can reflect on why they are following the hint. Understanding why their choice matters is crucial in actually learning rather than plainly following instructions. After the user has had enough time to read these cards and understand the reasoning, the cards fade out so that they can now return and focus on the task at hand.

Once the extinguisher is interacted with, it is equipped in the user's hand in a first-person view, and they are able to walk around the room while holding it. By pressing and holding the input button on the handheld controller, the extinguisher releases spray particles and a hissing sound that aims to represent the extinguishing agent being expelled from the nozzle. Both the fire and

the spray use colliders so that the system can detect when the user is actually aiming at the fire. The fire is programmed to extinguish when the extinguisher is sprayed on the fire continuously for roughly four seconds.

This same interaction logic is used in both the office and the kitchen scenes. The only difference is the type of fire and subsequently the correct extinguisher. In the office, the fire represents an electrical hazard which is a Class E fire but can easily spread and cause Class A materials to catch on fire, whereas in the kitchen it represents an oil fire which is a Class F type, replicating the differences discussed earlier regarding fire safety. The main purpose is to demonstrate a basic but complete chain of actions that mimic real world scenarios, such as noticing the fire, activating the alarm, choosing the right extinguisher, understanding why the extinguisher choice is correct or wrong, and finally extinguishing the fire.

5.2 VR System Setup and Interaction Framework

The simulation is built around a virtual reality interaction system so that users can move through the environment and interact with. The project uses Unity's XR Interaction Toolkit along with OpenXR, which provides head tracking, controller input, and object interaction for modern VR headsets. The VR rig, known as the XR Origin, is placed directly inside the office scene and kitchen scene and acts as the user's body in the virtual environment.

The XR Origin contains the main camera, which follows the user's head movements, and two controllers, right and left, that represent the hands. These are tracked in real time since the controllers are linked to the headset, so when the user raises their hand and points at something, the motion is shown inside the scene as well. The XR toolkit performs this tracking automatically.

Movement inside the simulation is handled using joystick-based movement instead of physical walking. A CharacterController component is attached to the XR Origin, and a Continuous Move Provider reads the input from the left controller's joystick. Pushing the joystick forward makes the player move in the direction they are looking, while the CharacterController ensures that the player cannot move through walls, furniture, or other obstacles. This setup reduces the risk of the users colliding with objects in the real world, since they do not have to physically walk to move around the simulation.

Some objects in the scene can be interacted with using XR components so they can be picked up and used directly in VR. The fire extinguishers are configured as XR Grab Interactables with colliders, rigidbodies, spray particle systems, and audio sources. Each extinguisher also has a script that tracks whether it is currently held. When the user grabs an extinguisher using the controller, XR events call a simple function to mark it as being held, which enables the spraying logic. Releasing the extinguisher disables this state again. The spray is activated through an input action mapped to the controller's trigger, and the raycast that checks whether it hit the fire which is cast from the nozzle on the extinguisher. XR takes care of grabbing and input actions, and the extinguisher script handles spraying and communication with the fire controller.

The fire alarm button follows a similar pattern. It has a collider and an XR Simple Interactable component, along with a script that controls the alarm sound. When a user interacts with the button using the controller, the XR event system calls the alarm's method. This starts the siren audio and also acts as a signal for the rest of the scenario logic, which is described later in Section 5.6.

On top of the basic interactions, there is a simple hint and feedback layer that supports the training aspect of the simulation. This implemented through small scripts that can enable or disable highlights and instruction cards. The correct extinguisher will have an outline and the information cards are pre-placed on the wall but remain inactive until they are needed. When the scenario logic decides that a hint should be shown, it calls these scripts to activate the highlight and briefly display the cards. The exact timing and conditions under which this happens are covered in more detail later on.

The same VR framework is used in both the office and kitchen scenes. In both cases, the user moves using the joystick, is able to look around using head tracking, interact with the alarm through XR events, and can pick up extinguishers via the grab interactable system. The only differences are in the environment layout, the type of fire being represented, and the configuration that decides what extinguisher type is treated as the correct option. This structure makes it easier to reuse and maintain the system, while still allowing each scene to represent different fire hazards and extinguishers.

Overall, the VR interaction framework is designed to stay easy to understand. The XR Origin manages tracking and locomotion, XR components handle grabbing and activation, and small scripts linked to individual objects manage behaviour and feedback. This approach allows the

technical design to be straightforward enough to extend the system with additional scenarios or more advanced interactions in the future.

5.3 3D Modelling and Environment Development

5.3.1 Modelling Workflow

The modelling for this simulation was carried out in Blender, and the focus was on creating simple, low-poly assets that were easy to read and easy to run inside VR without overloading it. The goal was not to build highly detailed rooms, but to design environments with the key elements of the fire scenarios.

The structural parts of both rooms were created first. This included the walls, floors, ceiling panels, door frames, and the overall layout that defines the environment itself. These shapes were kept simple, since the main purpose of the rooms is to provide a clean space where the user can walk, look around, and interact with the fire-safety equipment. Additional objects such as cabinets, shelves, tables, countertops, and most importantly the fire-alarm button were modelled using the same approach. Each asset was shaped to look familiar without adding unnecessary detail that might distract the user from the training.

The extinguishers were also modelled from scratch, and these required slightly more attention to detail as compared to the room furniture because they play a direct role in the user's interaction. Four types of extinguishers were created: water, foam, carbon dioxide, and dry powder. This was done to match the common extinguisher categories usually found in such scenarios, as discussed earlier in the thesis. These models include the main cylinder body, the handle, and a hose or nozzle depending on the extinguisher type. Along with the different design style based on each extinguisher, there are labels on each, so that users can visually distinguish the extinguishers from one another.

Figure 16: Model of CO_2 Extinguisher made in Blender

Figure 17: Model of Water Extinguisher made in Blender

Figure18: Model of Foam Extinguisher made in Blender

Figure 19:Model of Dry Powder Extinguisher made in Blender

Throughout the modelling process each asset was created to contribute to the fire-safety scenario in some way, either the environment the user walks on or part of the equipment the user interacts with. By keeping the workflow simple, it became easier to update or recreate objects whenever the rooms needed small adjustments.

5.3.2 Unity Integration

Once the models were prepared in Blender, the next step was to bring them into Unity and assemble the final environments. All assets were exported as FBX files and imported through Unity's Project window, where they could be organized into clear folders for each room. The purpose of this stage was simply to place the objects correctly, adjust their proportions, and prepare the environment so it behaves like an actual room when the user moves through it in VR.

The first set of adjustments focused on scale. Blender and Unity sometimes interpret object size differently, so a few models appeared much larger or smaller than expected when first imported. These were resized directly in the Inspector until their proportions matched practical real-world sizes. This was done using the VR camera inside the scene as a reference to judge whether a table was too tall, a cabinet was too wide, or an extinguisher looked oversized compared to the rest of the room.

After correcting the scale, each object was placed in its position within the scene. Furniture, shelves, wall units, kitchen counters and other environmental props were arranged to match the intended layout of the office and kitchen rooms. The goal was to create clear layout for the user to walk to and to interact with the key training elements in locations that make sense. Both rooms were kept relatively simple so that the user can easily identify the extinguishers, the fire alarm, and the fire itself without visual clutter.

Colliders were added to all structural and environmental objects to make sure the user cannot walk through walls, clip through furniture, or fall through the floor. For larger objects like walls or cabinets, simple box colliders were sufficient. Only objects that the user is not supposed to interact with physically received colliders at this stage since interactive behaviour is handled separately.

The objects were placed everywhere and had basic materials assigned to them. Since the modelling workflow relied on simple surfaces rather than detailed textures, most materials consisted of flat colours or basic Unity shaders. This choice kept the scene easy to understand and avoided unnecessary distractions during VR use. The emphasis was on clarity rather than realism, so colour choices were made to create good contrast between the environment, the extinguishers, and the fire-related elements.

Lighting in both scenes was kept minimal. Only a few directional or point lights were added to give the rooms a consistent and comfortable level of brightness. These lights were placed to illuminate the general environment rather than to simulate complex lighting behaviour. More advanced or dynamic lighting effects are handled later by the fire and particle systems, which are discussed in the next section. Here, the focus was only on giving the scene enough visibility so that the user can identify objects clearly once they load into the room.

To keep it organized, each room was assembled inside its own Unity scene file, with a clear hierarchy that separates structural elements, props, and interactive objects. This makes the layout easier to manage and reduces confusion when modifying the environment later.

5.4 Fire, Smoke, Lighting, and Audio Systems

5.4.1 Fire Particle System

The fire in both scenes is created using two overlapping flame particle systems placed on the same object to form a fuller flame shape. These particle systems loop continuously and act as the main indicator of an active hazard. A collider is placed inside the fire object so that the extinguisher's raycast has a consistent surface to detect.

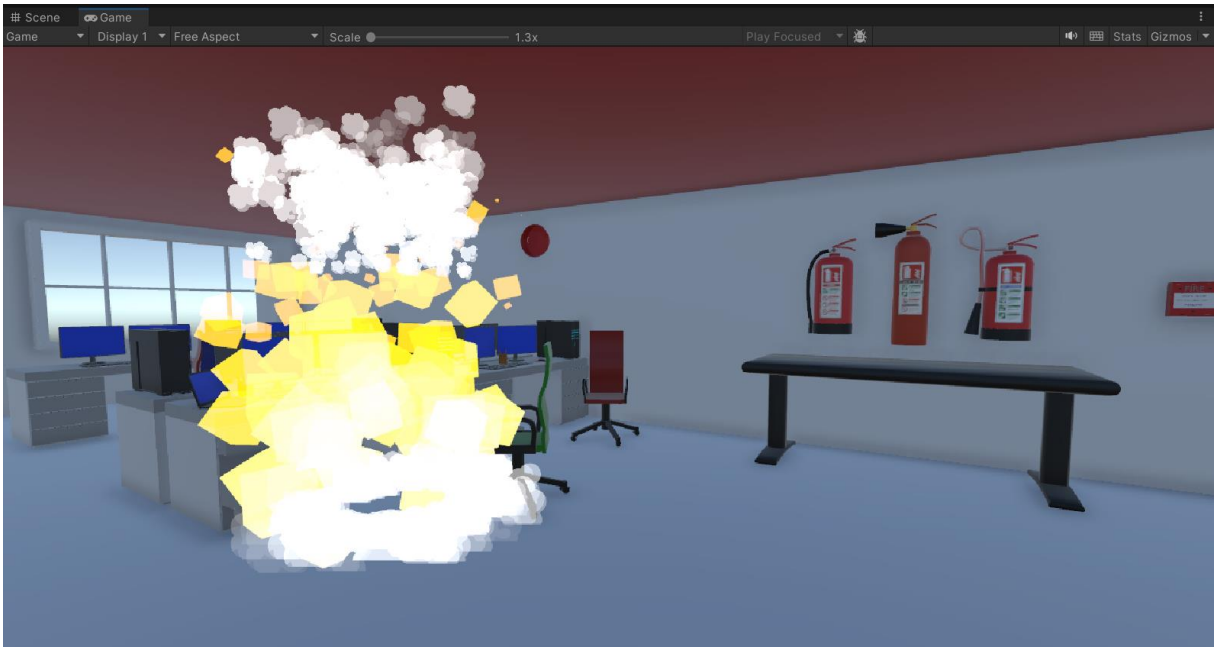

Figure 20: Simulated Fire Behaviour in the VR Office Scene

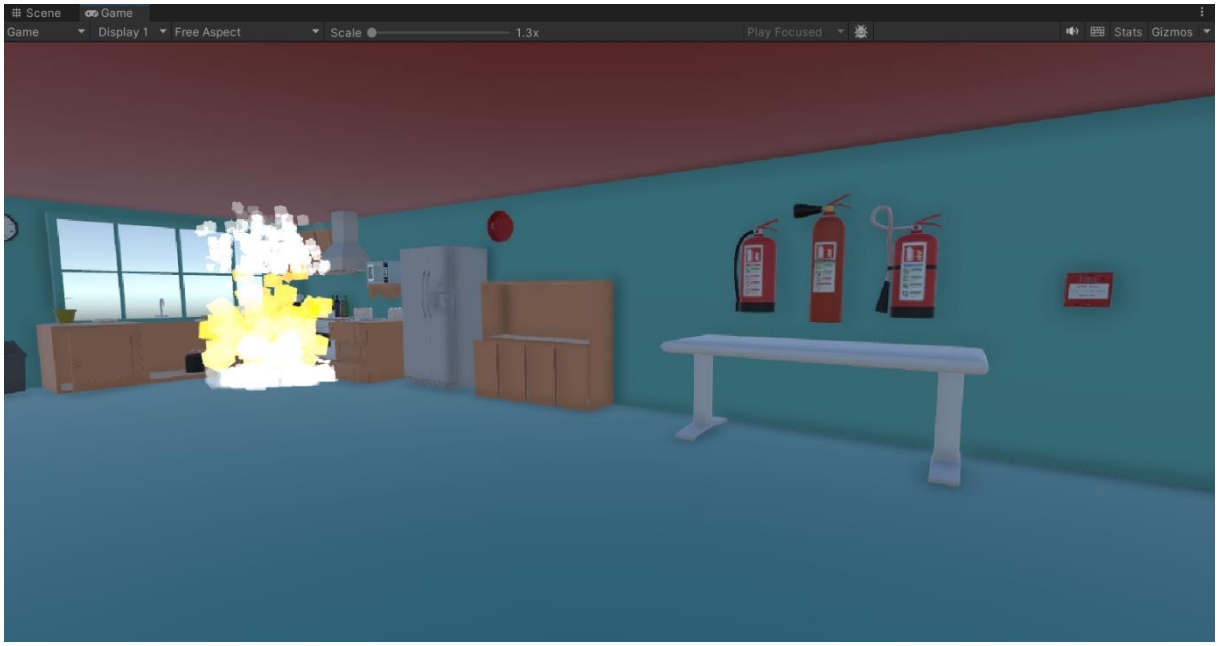

Figure21: Simulated Fire Behaviour in the VR Kitchen Scene

When the extinguisher spray hits this collider, the FireController script records how long the contact is maintained. After roughly four seconds of continuous spray, the fire begins its extinguishing sequence. Instead of deleting or disabling the fire object, the script gradually reduces the emission rate of all flame particles and then stops them entirely. This keeps the logic clean and allows the fire to be reused if the simulation is expanded later.

5.4.2 Smoke and Extinguisher Spray

The smoke used in the scenes comes from an additional particle system grouped under the same GameObject as the fire. The smoke also loops and stops at the exact moment the fire is extinguished. It fades out through the same FireController script, ensuring that both the flame and the smoke shut down together.

The extinguisher uses a separate particle system that produces a straight-line spray effect from the nozzle. This spray activates only while the VR trigger is held and stops instantly when released. Its purpose is functional rather than physical, the spray does not collide with the fire particles themselves. Instead, the extinguisher uses a raycast directed from the nozzle to determine whether the user is aiming correctly at the fire's collider. As long as the raycast hits, the extinguishing timer continues.

5.4.3 Lighting

Lighting in the scenes is kept simple. Unity's default directional light is used to give the rooms basic visibility, and no additional point lights or glow effects were added. The fire's visual glow comes entirely from the emissive materials and settings included in the particle assets themselves. This reduces unnecessary complexity and keeps the environment clear while still letting the fire stand out visually.

5.4.4 Audio Cues

Three looping audio sources support the particle effects.

- The fire object plays a crackling loop for as long as the flame particles are active.
- The extinguisher has a hissing loop that starts the moment the spray activates and stops as soon as the trigger is released.
- The fire alarm uses a continuous siren sound that begins when the alarm is activated and continues until the simulation ends.

Each sound is attached to its corresponding object through standard Unity AudioSources, and the FireController handles fading the fire audio out during the extinguishing sequence.

5.5 Interaction Mechanics

5.5.1 Fire Extinguisher Mechanics

The fire extinguisher is the main tool the user interacts with in the simulation, so its behaviour is kept simple and predictable. The extinguishers are mounted on hooks along the wall, and the user can walk close to them and reach out with the VR controller to pick one up. When the user grabs an extinguisher, it snaps into their hand in a stable orientation so that it feels like they are actually holding it, rather than floating awkwardly in front of them.

Figure 22: Fire Extinguishers Mounted on the Wall Inside the VR Environment

Once an extinguisher is equipped, the user operates it using the trigger on the controller. Pressing and holding the trigger releases a straight spray from the nozzle, and the system only sprays for as long as the trigger is held down. Releasing the trigger stops the spray immediately. The direction of the nozzle determines where the spray is aimed, so the user must physically point the extinguisher toward the fire. This encourages the same behaviour that is expected in real life, where the user needs to aim at the base of the flames rather than waving the extinguisher randomly.

The extinguishing mechanic depends on keeping the spray on target. While the user is aiming correctly at the fire, the system counts how long the spray has been hitting it. The fire is only put out after roughly four seconds of continuous spray. If the user moves the extinguisher away or stops spraying halfway, this contact is interrupted and the fire does not go out.

To guide the user, a highlighting feature is included. After the alarm has been activated, if no extinguisher is interacted with within a short time, the correct extinguisher for that scenario becomes visually highlighted. This does not force the user to take it, but it acts as a subtle hint that draws their attention to the appropriate choice. Once the user picks up the extinguisher, the highlight is no longer needed and can be removed so that the focus shifts back to the act of putting out the fire.

5.5.2 Fire Alarm Button

The fire alarm button is designed to be one of the first interactions the user performs in the simulation. It is placed in a visible and reachable location inside the room so that the user can approach it naturally. To activate it, the user only needs to move close to the button and perform a simple press action with the controller. There is no need for complex gestures or additional steps; one deliberate action is enough to trigger the alarm.

Figure 23: Fire Alarm Model Created in Blender

The moment the alarm is activated, a continuous siren sound begins to play. This gives immediate feedback that the interaction was successful and reinforces the idea that raising the alarm is a critical part of responding to a fire. The alarm does not need to be reset or acknowledged again during the simulation. It remains active in the background, reminding the user that the situation is still going on and that further decisions such as selecting and using the correct extinguisher is still required.

The button itself does not move or animate in a complicated way. The focus is on the cause-and-effect relationship: the user presses the alarm, and the system reacts instantly. This mirrors the behaviour of real manual call points, where the priority is to trigger the alarm as quickly as possible, not to provide a detailed physical response from the device itself.

5.5.3 Locomotion

Movement in the simulation is handled through continuous locomotion. The user moves by using the joystick on the VR controller, which allows them to walk forward, backward, or sideways at a controlled pace. The direction of movement is based on where they are looking, so turning the head changes the direction in which the joystick will move them. This makes navigation feel natural, because the user simply looks where they want to go and then moves in that direction.

The user is free to move around the room, approach the fire from different angles, or step back to create distance when needed. At the same time, movement is constrained by the physical layout of the scene. Turning is handled by the user's real body which helps reduce motion discomfort.

Overall, this locomotion method gives the user enough control to explore the environment and position themselves correctly for each interaction, while still keeping the controls simple enough to learn quickly. The user can focus on deciding where to stand, how close to get to the fire, and how to approach the extinguishers and alarm button, without having to manage a complicated movement system at the same time.

5.6 Scenario Logic

The scenario logic manages the order in which the events unfold and makes sure the simulation behaves consistently every time it is used. When the scene begins, the fire is already active, and the system starts in a "waiting for user response" state. Nothing progresses automatically until the user interacts with the environment, which helps the simulation feel predictable and allows the user to decide how they want to approach the situation.

The first significant change in the scenario occurs when the fire alarm is activated. Once the alarm button is pressed, the system marks this step as complete and begins monitoring the user's behaviour around the extinguishers. If the user does not interact with any extinguisher after a short delay, the correct extinguisher for that fire type is highlighted. This hint does not interfere with the user's choices, but it is a reminder that reacting quickly and choosing the right equipment can make a big difference in being able to control the incident. The highlight only remains until the correct extinguisher is picked up, after that it goes away to bring the user's attention back to the task.

When the extinguisher is equipped, the next phase of the scenario begins. At this point, three short information cards appear above the extinguisher rack. These cards explain why the selected extinguisher is the correct one for that specific fire and why the other two options wouldn't be effective or would be unsafe. The goal here is to give the user a small moment to understand the reasoning behind their decision. The cards stay visible long enough for the user to read them, and then they fade out automatically so the user can continue focusing on extinguishing the fire.

The final stage of the scenario revolves around the act of putting out the fire. The system monitors whether the extinguisher spray is aimed correctly and keeps track of how long the fire is being sprayed continuously. The fire will only be extinguished once the spray has been kept on target for the required duration. This prevents the user from extinguishing the fire by accident or by briefly tapping the trigger, and instead encourages steady and deliberate aiming. Once the fire is put out, the smoke, particles and audio are stopped immediately, and the scenario reaches its concluding state.

Overall, the logic is built to guide the user through a realistic sequence of actions without forcing them through a specific path. Each step reacts to the user's choices, and the system only moves forward when the user completes a task. This creates a simple but complete learning experience that reflects how a basic fire emergency situation would unfold.

5.7 Summary

This chapter presented the complete technical structure of the VR fire safety simulation and explained how all the major components work together inside the system. Each part of the environment, from the extinguishers and alarm button to the fire behaviour, particle effects and locomotion, was implemented in a way that keeps the interaction simple while still reflecting how basic fire response would unfold in real life. The scenario logic connects these elements into a single sequence so the user can move through the room, recognise the fire, activate the alarm and select the correct extinguisher before putting the fire out.

6. CONCLUSION

6.1 Summary

This project set out to build a virtual reality simulation that walks a user through the necessary steps of responding to a small fire incident. To guide the user through a simple but complete sequence of noticing the fire, activating the alarm, picking up an extinguisher, and using it correctly until the flames are fully extinguished. The simulation is not meant to replace professional training, but it gives a clear demonstration of what a basic response should look like in a controlled way.

The main strength of the system is the way the interactions are linked together. When the user pulls the alarm and then actually has to hold and aim the extinguisher themselves, it turns the experience from just watching a scenario to actively having to do it step by step. Most design choices were made to keep the procedure easy to understand. This helps keep the user's attention on the order of actions, which is the main goal in a training-style simulation.

Overall, the project reached its objective of creating a small, focused VR environment that demonstrates the core actions for fire-response. Even with simple models and mechanics, the simulation shows how VR can be used to rehearse safety procedures in situations where real practice would be risky and difficult to arrange.

6.2 Technical Reflections and Design Insights

Learning Blender came surprisingly quickly, mostly because I prefer figuring things out by actually doing them and after a few weeks of experimenting with modelling, rotating objects and navigating the 3D space, I reached a point where most of the workflow felt natural. One issue I faced during export was related to materials. Sometimes Unity would disable the material names and the colours would come out wrong, so I had to dig through the directories to reconnect the missing materials and make sure each asset matched the original planned design. Apart from that, the modelling and import process was one of the smoother parts of the project.

The interaction systems inside Unity however needed a bit more trial and error. Collision among objects was far more unpredictable because both raycast and collider setups kept giving errors or not registering hits properly at first and some objects even fell through the environment or

went straight through walls, and the same happened with the fire and smoke until I realised the room needed accurate colliders for the entire simulation to hold together.

Keeping the project low poly helped a lot, especially since I had to test and run everything on normal hardware, not a high-end machine. From a distance the environment looks clean but up close you still see the sharp edges that come with low poly assets, which is acceptable when performance matters.

With my background in computer science engineering, the coding aspect of everything came naturally so the logic itself was not difficult. The challenge was learning how that logic connects to actual game objects in Unity like attaching scripts, adjusting object parameters, and tweaking settings until they feel consistent.

Out of everything, the biggest change was how much this project improved my confidence when it comes to 3D modelling and creating whole scenes. Blender and Unity went from something I was curious about to now something I can use comfortably to build whatever I imagine. Converting an idea into a physical 3D object and then seeing it come alive inside a VR simulation, is a skill I gained from this project, and it ended up being one of the most valuable parts of the entire experience.

6.3 Future Development Directions

The current simulation focuses on single, contained rooms, but future development would be to expand into environments that feel closer to real life. Server rooms, chemical labs, and residential building layouts are some of the scenarios that stand out the most, not only because they introduce new hazards, but because they mirror the types of fires that people might encounter outside just the controlled VR setting. A setup where there are multiple rooms connected some areas are stable, others have hidden risks, and a few are active hazards would make the experience feel more unpredictable and realistic.

One very meaningful upgrade, although one of the most challenging would be adding randomised fire behaviour. Real fires are very unpredictable and recreating their behaviours in VR requires more than coding a simple effect. Features like flames spreading around, smoke density changing based on other factors, and conditions that worsen when the wrong extinguisher is used all contribute to that ideal realism. Even allowing a fire to reignite if the

user sprays at the fire inaccurately, or takes too long to react to it, would make future scenarios feel closer to the pressure of real emergencies.

Pulling the safety pin before using an extinguisher, getting haptic feedback while grabbing a door handle only to realise it is hot, or even reacting to that same haptic feedback when touching something unsafe are all small actions that immerse the user better. Introducing techniques such as the PASS method, followed by test modes with random situations, would help users first learn the steps and then prove that they can repeat them without any assistance. In the long term, adding an analysis layer can make the simulation into a proper training tool that tracks performance. Tracking how fast the user detects a fire, how well they handle the extinguisher itself, or whether they triggered the alarm at the right time allows the system to give meaningful feedback. With enough attempts recorded, this data could even reveal common mistakes made by different groups of people, which would be valuable for researchers trying to improve safety training.

Multi-user drills where students or staff are part of the same environment together could help encourage communication and teamwork skills.

In a more advanced version of the simulation, they could recreate real fire incidents from past cases so that trained firefighters can study what went wrong and attempt different strategies to achieve better outcomes this time. This type of repetition in a safe environment has real value in these type of scenarios which can’t realistically be recreated in real life.

Finally, Mixed reality in particular could turn the user’s real surroundings into part of the training, getting rid of the need for large empty spaces. This direction represents one of the most promising ways to make the simulation realistically intense by making the danger feel closer to the physical world.

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